ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician (MAR TECH) 00379

USQ-MS SENIOR ELECTRICAL SPECIALIST (ALUZ)

MITE Course ID: 122021

RECORD OF AMENDMENTS

1. This Qualification Standard and Plan (QSP) is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0 (QSPWB May 18)		NTDC(P) CO
	17/06/2022 (Piloting)	Sebastien Auger, HOD COPS, NPTG HQ
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FOREWORD

1. This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-002 MAR TECH MOS ID 00379 dated 28 Apr 17;

c. NAVORD 4500-0 “Structured Delegation”; and

d. NPTORD 4500-22 “Qualification Standard and Plan Production and Management”.

2. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the Curriculum Control Authority (CCA) (NTDC(P) MSE Division), who will forward to NPTG for approval, as appropriate.

5. The designated Training Establishments (TEs) for this training are indicated below:

TE	Phase / Module	MITE Course ID	NQual
Naval Fleet School (Pacific) (NFS(P))	All	122021	ALUZ
Naval Fleet School (Atlantic) (NFS(A))	All	122021	ALUZ

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL TRAINING GROUP (CNTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Senior Electrical Specialist (Sr Elec Spec). This QSP is designed to assist the training system in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP.

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points	26,220
Enabling Checks (ECs)	2,820
Performance Checks (PCs)	2,825
ATT Sub-total	**31,865**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	1,380
Training Administrative Time (TAT)	1,380
Training Travel Time (TTT)	0
Physical Training (PT)	3,420
Sub-total	**6180**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**95**

RESOURCES

3. The following table details key personnel required to conduct the training:

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Sr Instructor	PO1	MAR TECH RQ-PO1 (ALUV)	1	001-009
Instructor (NTI)	Civ	Electrical engineer (or certified technologist)	2	001-004
Subject Matter Expert (SME)	MS – PO1	USQ-MS Sr Elec Spec (ALUZ); Senior Electrician posted onboard MCDV	1	002.02 002.03 002.04

Instructor	MS / PO2	USQ-MS Sr Elec Spec (ALUZ) with experience in Controls Technician position onboard ship	2	005-009
SME lecturer	Civ	Field Services Representative	1	005.04

4. The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP. Details as to required equipment in labs and trainers is found at Annex H.

Critical Training Resources	PO/EO
Stationery	All
Scientific calculator (on loan)	All
General classroom	All
Textbooks (on loan)	001 – 004
Test equipment	All
Lab Volt equipment	002, 004
Marine Coastal Defence Vessel (MCDV) (electrical tour)	002.04
Local Area Network (LAN) / Network lab	003
Closed Circuit Television (CCTV) server and cameras	003.03, 005.03
Programmable Intelligent Computer (PIC) trainer with Lab Volt interface	004
Tool kit	004
Equipment Health Monitoring (EHM) Server	005.03
Platform Control Operator Trainer (PCOT)	005 – 009
Integrated Platform Management System (IPMS) Maintenance Trainer	005 – 009

COURSE CAPACITY

5. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Complete justification can be found in Annex C.

a. Maximum – 8 (resource limitation of Lab Volt Trainers);

b. Minimum – 4 (less than 4 would have limited return on investment due to instructor time constraints); and

c. Optimum – 8 (to maximize resources).

MITE

6. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

7. Scheduling. Each session of this course shall be scheduled in accordance with (IAW) NPTORDs 4540-0 - Quantity Control Process for Individual Training and Education, 4540-2 - RCN Course Scheduling, and 4540-43-Training Event Management. It is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in Military individual training and education (MITE) is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays.

8. Prerequisites. The loading authority must ensure that trainees have the following prerequisites before being selected for this course:

a. Unique Specialty Qualification – Sailor 1st Class (USQ-S1) Electrical Specialist (ALUT); and

9. Qualification: Successful completion of this course constitutes the primary means of attaining the MAR TECH USQ-MS Senior Electrical Specialist (ALUZ, MITE 122021).

COURSE REPORTING

10. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51-Course Reports and Certificates.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

11. PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces Military Equivalencies Program”, and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment”.

SPECIAL INSTRUCTIONS

12. Course is geared toward senior S1 and newly-promoted MS.

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE CCA NFS (P)

GENERAL

1. This is a pass/fail course. Trainees must successfully complete all POs to pass this course. Performance Checks (PCs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

2. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K for details.

3. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

ENABLING AND PERFORMANCE CHECKS

4. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

5. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES Vol 6 with the unsatisfactory progress documented in their Trainee File.

6. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

7. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

8. All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 “Training Review Board Process”. Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

RESPONSIBILITY FOR TESTING

9. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners”.

10. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

PROGRESS MONITORING

11. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.

12. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

13. Continuous monitoring of candidate progress is required to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for any required TRBs; and

d. Information to assist in completion of the CF 377 Course Report.

14. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

15. Instructional staff shall maintain individual trainee files, including the results of all:

a. Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);

b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and

c. Summative assessment material (PCs).

TRAINING REVIEW BOARD

16. TRBs shall be convened and conducted IAW with NPTORD 4500-54 “Training Review Board Process”.

17. In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis, the necessity to convene a TRB.

COURSE REPORTS

18. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51 “Course Reports and Certifications”.

19. Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

20. The Course Report narrative comments could include:

a. Ability to work with others;

b. Ability to work under pressure;

c. Written and verbal communication skills;

d. Organizational skills; and

e. Leadership ability.

21. Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE CCA NFS (P)

PO 001 – Interpret Circuit Fundamentals

1. Performance. Interpret Circuit Fundamentals.

2. Conditions.

   a. Given:

      (1) HMC Ship;

      (2) Test equipment; and

      (3) References;

   b. Denied: Nil.

   c. Environmental:

      (1) Ship at sea or alongside; and

      (2) Any environmental conditions.

3. Standard. The Senior Electrical Specialist shall interpret circuit fundamentals IAW refs by:

   a. Analyzing digital circuit fundamentals; and

   b. Analyzing analog circuit fundamentals.

4. References. C116, C117.

5. Task, Skill, Knowledge Numbers. Nil.

6. Training Limitations.

   a. PO 001 incorporates foundational theory and skills required for Senior Electrical Specialists to troubleshoot networks and network components therefore this PO is to be instructed before all others in this QSP.

PO 001 – Interpret Circuit Fundamentals

EO 001.01 – Analyze Digital Circuit Fundamentals

1. Performance. Analyze Digital Circuit Fundamentals.
2. Conditions.
   a. Given:
      (1) Stationery; and
      (2) Calculator.
   b. Denied:
      (1) Assistance; and
      (2) References.
   c. Environmental: Classroom.
3. Standard. The trainee shall analyze digital circuit fundamentals IAW refs by:
   a. Analyzing number systems;
   b. Analyzing number system conversions;
   c. Analyzing digital logic fundamentals; and
   d. Analyzing digital circuits.
4. Time Required.
   a. EO – 900 mins to deliver training; and
   b. EC – 70 mins (total for all trainees).
5. Remarks. This EO shall be taught as the first phase of the course as it will be required for follow-on POs.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Analyze number systems			
1. Analyze: a. Decimal numbering system. b. Binary number system. c. Hexadecimal system.	C116, C117	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will explain teaching points. Trainees will complete worksheets in class based on the teaching points. Worksheets will be self-assessed based on a standardized solution / answer sheet. **Trg Resources:** • Worksheets	50 mins

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Standardized solution / answer key	
LESSON OBJECTIVE: Convert number systems			
1. Converting: a. from decimal to binary; b. from decimal to hexadecimal; c. from binary to hexadecimal; d. from hexadecimal to decimal; and e. from binary to decimal.	C116, C117	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will explain teaching points. Trainees will complete worksheets in class based on the teaching points. Worksheets will be self-assessed based on a standardized solution / answer sheet. **Trg Resources:** • Worksheets • Standardized solution / answer key	50 min
LESSON OBJECTIVE: Analyze digital logic fundamentals			
1. Analyze: a. AND truth tables and timing diagrams; b. OR truth tables and timing diagrams c. NAND truth tables and timing diagrams; d. NOR truth tables and timing diagrams; e. XOR truth tables and timing diagrams; f. XNOR truth tables and timing diagrams; g. FLIP FLOP truth tables and timing diagrams to include: (1) Set/Reset; (2) D type; and (3) JK type.	C116, C117	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will explain teaching points. Trainees will complete worksheets in class based on the teaching points. Worksheets will be self-assessed based on a standardized solution / answer sheet. **Trg Resources:** • Worksheets • Standardized solution / answer key	300
LESSON OBJECTIVE: Analyse digital circuits			
1. Utilize: a. Flip/flop asynchronous inputs; b. Flip/flop synchronous inputs; c. Asynchronous ripple counter; d. Johnson counter;	C116, C117	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will explain teaching points. Trainees will complete	500

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
e. Shift registers; f. Multiplexers; and g. Decoders.		worksheets in class based on the teaching points. Worksheets will be self-assessed based on a standardized solution / answer sheet. **Trg Resources:** • Worksheets • Standardized solution / answer key	

7. Test Details. EO 001.01 will be tested in EC 001.01 IAW table below and Annex K. The material will be summatively assessed throughout follow-on POs.

EC 001.01: Analyze digital circuit fundamentals

EC Time: 60 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will complete a written test and submit for assessor feedback based on standardized solutions. Trainees will not have access to assistance or references for the EC.
Pass/fail criteria: 70% pass mark required.

ASSESSMENT RESOURCES:
- Nil

IMPACT OF FAILURE: Remedial Training / Re-test

PO 001 – Interpret Circuit Fundamentals

EO 001.02 – Analyze Analog Circuit Fundamentals

1. Performance. Analyze Analog Circuit Fundamentals.

2. Conditions.

   a. Given: Stationery.

   b. Denied:

      (1) Assistance; and

      (2) References.

   c. Environmental: Classroom.

3. Standard. The trainee shall analyze analog circuit fundamentals IAW refs by:

   a. Analyzing Op-amp circuit theory;

   b. Analyzing the operation of linear op-amp comparator circuits; and

   c. Analyzing the operation of non-linear op-amp comparator circuits.

4. Time Required.

   a. EO – 800 mins to deliver training; and

   b. EC – 70 mins (total for all trainees).

5. Remarks. This EO shall be taught as the first phase of the course (directly following or concurrent with EO 001.01) as it will be required for follow-on POs.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Analyze Op-amp circuit theory			
1. Apply basic rules for solving op-amp circuits; 2. Analyze inverting op-amp circuits; 3. Analyze non-inverting op-amp circuits; 4. Analyze voltage follower op-amp circuits; and 5. Analyze combination circuits.	C116, C117	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will explain teaching points. Trainees will complete worksheets in class based on the teaching points. Worksheets will be self-assessed based on a standardized solution / answer sheet. **Trg Resources:** • Worksheets • Standardized solution / answer key	200
LESSON OBJECTIVE: Analyze the operation of linear op-amp comparator circuits			
1. Analyze types of comparators:	C116,	**Method / Media:** F2F	300

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
a. Summing amplifier; b. Difference amplifier; and c. Instrumentation amplifier.	C117	**Trg Environment:** Classroom **Activity Details:** Instructor will explain teaching points. Trainees will complete worksheets in class based on the teaching points. Worksheet will include a schematic from an HMC Ship for trainees to identify the different types of amplifiers to show how Op-amp comparator circuits apply onboard ship. Worksheets will be self-assessed based on a standardized solution / answer sheet. **Trg Resources:** • Worksheets • Standardized solution / answer key	
LESSON OBJECTIVE: Analyze the operation of non-linear op-amp comparator circuits			
1. Analyze Integrator circuits; 2. Analyze Differentiator circuits; and 3. Analyze Comparator circuits to include: a. Limit detector; and b. Window comparator;	C116, C117	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will explain teaching points. Trainees will complete worksheets in class based on the teaching points. Worksheet will include a schematic from an HMC Ship for trainees to identify the different types of amplifiers to show how Op-amp comparator circuits apply onboard ship. Worksheets will be self-assessed based on a standardized solution / answer sheet. **Trg Resources:** • Worksheets • Standardized solution / answer key	300

7. Test Details. EO 001.02 will be tested in EC 001.02 IAW table below and Annex K. The material will be summatively assessed throughout follow-on POs.

EC 001.02: Analyze analog circuit fundamentals

EC Time: 60 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will complete a written test and submit for assessor feedback based on standardized solutions. Trainees will not have access to assistance or references for the EC.
Pass/fail criteria: 70% pass mark required.

ASSESSMENT RESOURCES:

- Nil

IMPACT OF FAILURE: Remedial Training / Re-test

PO 002 – Troubleshoot Analog Control Systems

1. Performance. Troubleshoot Analog Control Systems (Task# TBD).

2. Conditions.

   a. Given:

      (1) Precision measuring tools and test equipment;

      (2) HMC Ship;

      (3) References;

   b. Denied: Nil.

   c. Environmental:

      (1) On ship or ashore;

      (2) At sea or alongside; and

      (3) Any environmental conditions.

3. Standard. The Senior Electrical Specialist shall troubleshoot analog control systems using precision tools and test equipment IAW refs by:

   a. Analyzing elements of controls by performing power calculations;

   b. Analyzing control circuits by:

      (1) Calibrating control sensors (ALT0140) to include:

         (a) Temperature sensors (AKT0330);

         (b) Speed sensors;

         (c) Level sensors;

         (d) Strain sensors;

         (e) Photo sensors; and

         (f) Pressure sensors.

      (2) Analyzing PID;

      (3) Analyzing closed loop systems;

      (4) Analyzing open loop systems;

      (5) Performing trials on SCR propulsion cabinet (Task# TBD);

      (6) Calibrating power generation control system (AUT0580); and

      (7) Operating workshop calibration equipment.

   c. Analyzing Four Quadrant Control by:

      (1) Maintaining power distribution systems;

      (2) Maintaining electrical propulsion control systems; and

      (3) Maintaining the propulsion system.

d. Analyzing chopper circuits by;

(1) Maintaining power generation control systems and components; and

(2) Maintaining Propulsion control systems and components.

e. Analyzing inverter circuits by:

(1) Maintaining MCDV electrical ancillary equipment;

(2) Calibrating circuit cards (AUT0470); and

(3) Maintaining electrical components of MCDV steering systems.

4. References. A50, A51, A52, C116, C117, C118, C119

5. Task, Skill, Knowledge Numbers. AKT0330; ALT0140; ALT2130; AUT0470; AUT0560; AUT0580

6. Training Limitations.

a. Lab Volt resources IAW PIC setup are required for training.

b. Gas turbine distributed control system and alternator trainer IAW NFS(A) setup is required for training.

PO 002 – Troubleshoot Analog Control Systems

EO 002.01 – Analyze elements of control

1. Performance. Analyze elements of control.

2. Conditions.

   a. Given: Stationery.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall analyze elements of control by:

   a. Comparing limit to functional notation;

   b. Analyzing functional graphs;

   c. Utilizing limits of a function to explain derivatives;

   d. Utilizing the rules of derivatives to:

      (1) Find the rate of change of a function;

      (2) Sketch functions; and

      (3) Find the original function.

   e. Applying the rules for integrals to:

      (1) Find the indefinite integral of a given function; and

      (2) Find the definite integral of a given function.

   f. Analyzing statistical functions using derivatives and integrals; and

   g. Analyzing electrical functions using derivatives and integrals.

4. Time Required.

   a. EO – 1150 mins to deliver training; and

   b. EC – 60 mins (total for all trainees).

5. Remarks. EO 002.01 shall be instructed by a civilian SME (electrical engineer or technologist equivalent; specialist in calculus with RLC circuits).

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Define closed loop elements			
1. Proportional; 2. Integral; 3. Derivative;	A50, A51, A52	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:**	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
4. Set point; 5. Process lag; 6. Feedback; and 7. Following Error.		Instructor will teach closed loop control theory. Instructor will give real life examples of PID. Sketch a simple block diagram, identify SP, FB, error and their roles in a well-sorted control system. **Trg Resources:** Handouts on closed loop control system.	
LESSON OBJECTIVE: Compare limit notation to functional notation by:			
1. Defining what a limit is; 2. Using the graph of a continuous function to discuss how limit notation is similar to functional notation; and 3. Discussing how limit notation tells more about the function than functional notation when the graph is discontinuous.	A50, A51, A52	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:** Instructor will demonstrate using graphs of continuous functions. Then discuss the limit notations of the function. **Trg Resources:** Nil	50
LESSON OBJECTIVE: Analyze functional graphs			
1. Use analytical functions to describe how the function behaves as it approaches a given point vs how it behaves at the given point.	A50, A51, A52	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Use graph example from last lesson and illustrate the difference between a value at or near a discontinuous point. Trainees will be given questions to complete and compare to a standardized solution. **Trg Resources:** Nil	50
LESSON OBJECTIVE: Use the limit of a function to explain what a derivative is			
1. Using a quadratic function F(x)=x2 show how a limit can be used to determine the rate of change of the function; 2. Refer to linear functions, and use the limit notation to prove the rate of change of the linear functions; and 3. Compare the derivatives of the linear to the quadratic functions.	A50, A51, A52	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will use quadratic function and show the rate of change on a graph. Trainees will be given a take home assignment to practise graphing points. They will self-assess based on standardized solutions. **Trg Resources:** Assignment	100
LESSON OBJECTIVE: Use the rules of derivatives to find the rate of change of functions			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
1. Finding the derivatives of polynomials; 2. Using the power rule find the derivative of the function; 3. Finding the second, third and fourth derivative of a function; 4. Perform power calculations.	A50, A51, A52	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:** Instructors will be introducing the concept that derivatives are distributive. Instructor will demonstrate how to use the rules of derivatives with trainees performing calculations to practice. **Trg Resources:** Nil	100
LESSON OBJECTIVE: Use the rules of derivatives to sketch functions			
1. Determining the intercepts and the max/min points of the function; 2. Determining which of the points from 1 are maximums and which are minimums; and 3. Sketching the function.	A50, A51, A52	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:** Instructor will demonstrate points on a graph and find the critical points required. Trainees will be given assignment on finding min and max points. Trainees will be given an in-class assignment on using the rules of derivatives. Instructor will be available to answer questions. **Trg Resources:** Assignment	150 100 (assign ment)
LESSON OBJECTIVE: Use the derivative of a function to find the original function			
1. Work backwards to determine the original function; 2. Develop the rule used to determine the original function; and 3. Determine what is missing from the original function and why.	A50, A51, A52	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:** Demonstrate with an example of how to work backward to find the original function. Trainees will be given a take home assignment to practise taking the derivatives of a function. **Trg Resources:** Assignment	50
LESSON OBJECTIVE: Apply the rules for integrals			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
1. Find the indefinite integral of a given function; and 2. Find the definite integral of a given function.	A50, A51, A52	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:** Instructor will demonstrate how to find the derivative of a slope on the board. Instructor will demonstrate different sine waves and relate them to DC and RMS. Trainees will follow and practise with other examples. **Trg Resources:** Nil	150
LESSON OBJECTIVE: Analyzing statistical functions using derivatives and integrals			
1. Determining functional information using functional notation and limit notation; 2. Determining the rate of change of a function at given points; and 3. Determining the total area under the function.	A50, A51, A52	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:** Trainees will be completing in class exercises on limits and their applications, derivatives and their applications, and integrals and their applications. **Trg Resources:** • Handout (difference between DC and RMS) • Assignment	200
LESSON OBJECTIVE: Analyzing electrical functions using derivatives and integrals			
1. Determining the voltage drops in series RLC circuits after a given time interval; 2. Determining branch currents in parallel RLC circuits after a given time interval; and 3. Predicting the outputs of integrators and differentiator circuits after a given time interval when given the applied input signal.	A50, A51, A52	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:** Instructor will demonstrate voltage drops on an RLC circuit and solve for the drops. Trainees will follow along and complete practise questions. **Trg Resources:** Assignment	150

7. Test Details. EO 002.01 will be tested in EC 002.01 and PC 002 IAW table below and Annex K.

EC 002.01: Analyze elements of control

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: EC 002.01 will be a written assessment based on content from EO 002.01. Trainees will not have access to references or handouts for exams.
Pass/fail criteria: 70%

ASSESSMENT RESOURCES:

- NIL

IMPACT OF FAILURE: Remedial Training / Re-test

PO 002 – Troubleshoot Analog Control Systems

EO 002.02 – Analyze Control Circuits

1. Performance. Analyze control circuits.

2. Conditions.

   a. Given:

      (1) Handouts;

      (2) Assistance; and

      (3) References.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) Lab.

3. Standard. The trainee shall analyze control circuits IAW refs by:

   a. Calibrating control sensors (ALT0140);

   b. Controlling the actuators;

   c. Calibrating PID;

   d. Defining DC power supply connections for rectifiers;

   e. Comparing the characteristic curve of electronic switch devices;

   f. Describing the operation of the three phase controlled rectifier supplying:

      (1) A resistive load;

      (2) An inductive load; and

      (3) An active load.

   g. Describing PWM control of a DC motor.

4. Time Required.

   a. EO – 1900 mins to deliver training; and

   b. EC – 120 mins (total for all trainees).

5. Remarks. Instructor should have received a tour prior to instructing this EO from a SME (Sr Electrician posted onboard an MCDV) so they are able to speak confidently to the MCDV platform specifics.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Calibrate control sensors (ALT0140)			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
1. Define: a. Resistance Temperature Detector (RTD) (3 and 4 wire); b. Strain gauges; c. Thermocouples; d. Hall effect devices; e. Linear variable differential transformers (LVDT); and f. Encoder wheels. 2. Identify DC bridge circuits for use with sensors; 3. Identify procedures for nulling and calibrating sensors in bridge circuits; 4. Identify methods of determining the sensitivity of bridge circuits; 5. Calibrating Temperature sensors (AKT0330): a. Positive Temperature Coefficient thermistors; and b. Negative Temperature Coefficient thermistors. 6. Repair precision control equipment by troubleshooting precision control equipment.	Relg/Satori, Industrial Electronics, Pearson Education, Inc., 2006 (ISBN 0-13-206418-9)	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructors will explain all teaching points in class. In lab, instructor will demonstrate procedures; trainees will have an assignment to complete; the instructor will assess each trainee's lab for completeness. **Trg Resources:** • Handouts (Calibration Procedure) • Lab Volt equipment • Sensors • Meters	100 F2F 100 Lab
LESSON OBJECTIVE: Control the actuators			
1. Define how input sensor information is utilized by the controllers and how information is then used to control the actuators; 2. Review specific sensors used; 3. Review control actuators used; 4. Analyze electrical schematics to explain how control is established; 5. Relate to shipboard system (i.e. IPMS, MCDV); and 6. Discuss troubleshooting examples of industrial control experience from instructor and/or trainee experience.	Relg/Satori, Industrial Electronics, Pearson Education, Inc., 2006 (ISBN 0-13-206418-9)	**Method / Media:** F2F, GD **Trg Environment:** Classroom **Activity Details:** Instructor to go over teaching points. Trainees will have a discussion relate their experiences onboard ship to shipboard specific systems with the instructor asking questions to guide the discussion as required. **Trg Resources:** • Nil	100 F2F 100 GD
LESSON OBJECTIVE: Calibrate PID			
1. Explain the hydraulic PID control circuit; 2. Calibrate hydraulic PID set-up;		**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt	100 F2F 300 Lab

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
3. Troubleshoot hydraulic PID set-up; and 4. Calibrate PID temperature control circuit.		**Activity Details:** Instructor will explain PID control and then the trainees will complete a lab on PID. Trainees will have a worksheet that they will work through; the instructor will assess each trainee's lab for completeness. **Trg Resources:** • Assignment	
LESSON OBJECTIVE: Define DC power supply connections for single and three phase rectifiers			
1. Describe grounding in Power Supply Systems: a. Basics of grounding; b. Basics of bonding; c. Ungrounded Power Supply System; d. Solidly Grounded Power Supply System; and e. Other types of grounding for power supply systems. 2. Analyze DC Bridge rectifiers; 3. Describe component functions: a. Transformers; b. Diodes; c. Silicon Controlled Rectifiers (SCR); d. Metal-Oxide Semi-conductor field-effect transistor (MOSFETs); and e. Programmable Logic Controller (PLCs). 4. Describe how to detect faulty components: a. Visually; and b. Via measurements.	Wildi, Electrical Machines, Drives and Power Systems, 6th Ed, Pearson Education, Inc., 2006 (ISBN 0-13-177691-6)	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructor to cover all teaching points, including identifying common equipment grounding using visual aids (photos) with focus on safety and troubleshooting. In lab, instructor will demonstrate how to detect faulty equipment with trainees practicing. **Trg Resources:** • Pictures of transformers, diodes, SCRs, MOSFETs, PLCs	200
LESSON OBJECTIVE: Compare the characteristic curve of electronic switch devices			
1. Define the characteristic curve of: a. Bi-Polar Junction Transistor (BJT); b. Metal Oxide Semi-conductor field effect transistor (MOSFET); c. Silicone Control Rectifier (SCR); d. Triode for alternating current (TRIAC); and e. Insulated gate bipolar transistor (IGBT).		**Method / Media:** F2F, brainstorm **Trg Environment:** Classroom **Activity Details:** Instructors to visually demonstrate the different devices and how they are located in a circuit. Trainees are to brainstorm the pro/cons of each device and how they relate to shipboard systems. **Trg Resources:**	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
2. Discuss the pros/cons of each device		• Visual aids for devices	
LESSON OBJECTIVE: Discuss the operation of the three phase controlled rectifier supplying a resistive load			
1. Describe the component function; 2. Explain circuit operation; and 3. Explain circuit troubleshooting.		**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructors to explain operation to trainees. Trainees will be given a circuit and write out the steps required for troubleshoot that circuit. Trainees will submit their troubleshooting procedure for instructor feedback. **Trg Resources:** • Circuit troubleshooting assignment	100
LESSON OBJECTIVE: Discuss the operation of the three phase controlled rectifier supplying an inductive load			
1. Describe the component function; 2. Explain circuit operation; and 3. Explain circuit troubleshooting.		**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructor will explain all teaching points in the classroom. In the lab, instructor will demonstrate relevant procedures, confirming trainees understanding of 4 quadrant converter and Buck/boost chopper; trainees will have an assignment to complete; the instructor will assess each trainee's lab for completeness. **Trg Resources:** • Assignment	250
LESSON OBJECTIVE: Discuss the operation of the three phase controlled rectifier supplying an active load			
1. Describe the component function; 2. Explain circuit operation; and 3. Explain circuit troubleshooting.		**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructor will explain all teaching points in the classroom. In the lab, instructor will demonstrate relevant procedures; trainees will have an assignment to	150 F2F 150 lab

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		complete; the instructor will assess each trainee's lab for completeness. **Trg Resources:** • Lab resources	
LESSON OBJECTIVE: Describe Pulse Width Modulation (PWM) control of a DC Motor			
1. Explain different pulse wave modulation (PWM) schemes to include: a. Constant frequency; and b. Varying frequency.		**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details**: Instructor will explain all teaching points in the classroom. In the lab, instructor will demonstrate relevant procedures; trainees will have an assignment to complete; the instructor will assess each trainee's lab for completeness. **Trg Resources:** • Assignment	150

7. Test Details. EO 002.02 will be tested in EC 002.02 and PC 002 IAW table below and Annex K.

EC 002.02: Analyze control circuits

EC Time: 100 mins (concurrent) + 20 min debrief

ASSESSMENT ACTIVITY: EC 002.02 will be a written assessment based on content from EO 002.02. Trainees will not have access to references or handouts for the written EC.
Trainees will be practically assessed through the lesson labs. They will have access to references and handouts during labs.
Pass/fail criteria: 70% on written; pass on all labs

ASSESSMENT RESOURCES:
- Nil

IMPACT OF FAILURE: Remedial Training / Re-test

PO 002 – Troubleshoot Analog Control Systems

EO 002.03 – Analyze Four Quadrant Control

1. Performance. Analyze Four Quadrant Control.
2. Conditions.
   a. Given:
      (1) Handouts;
      (2) Assistance; and
      (3) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) Lab.
3. Standard. The trainee shall analyze four quadrant control IAW refs by:
   a. Explaining the four quadrant control theory;
   b. Troubleshooting four quadrant control systems;
   c. Calibrating power generation control systems (AUT0580);
   d. Repairing MCDV power distribution systems;
   e. Adjusting MCDV electrical propulsion systems;
   f. Adjusting MCDV electrical control propulsion systems;
   g. Inspecting MCDV electrical propulsion systems;
   h. Troubleshooting MCDV propulsion control system (ALT2130); and
   i. Troubleshooting MCDV electrical propulsion system and controls.
4. Time Required.
   a. EO – 750 mins to deliver training; and
   b. EC – 60 mins (total for all trainees).
5. Remarks. Instructor should have received a tour prior to instructing this EO from a SME (Sr Electrician posted onboard an MCDV) so they are able to speak confidently to the MCDV platform specifics.
6. Teaching Points.

Teaching Points	Ref	Training Activity Details	Time (mins)
LESSON OBJECTIVE: Explain the 4 quadrant control theory			
1. Describe one quadrant control including variable voltage speed drive;	A50, A51,	**Method / Media:** EDI	450

Teaching Points	Ref	Training Activity Details	Time (mins)
2. Describe two quadrant control including regenerative braking effects; and 3. Describe four quadrant control including bi-directional operation, component function and verifying proper operation or failed operation through measurement.	A52, C116, C117, C118, C119	**Trg Environment:** Classroom; Lab Volt **Activity Details:** The instructor is to explain 4 quadrant control theory, relating theory to naval examples. Trainees will complete a lab on 4 quadrant control. **Trg Resources:** • Handout MCDV Drive system	
LESSON OBJECTIVE: Analyze four quadrant control			
1. Troubleshoot 4 quadrant control system; 2. Calibrate power generation control system (AUT0580); 3. Repair power distribution system (MCDV); 4. Adjust electrical propulsion system (MCDC); 5. Adjust electrical control propulsion system (MCDC); 6. Inspect electrical propulsion system (MCDV); 7. Troubleshoot operating irregularities of the propulsion control system (MCDV); and 8. Troubleshoot electrical propulsion system and controls (MCDV).	A50, A51, A52, C116, C117, C118, C119	**Method / Media:** Lab **Trg Environment:** Lab Volt **Activity Details:** Instructor to aide trainees in lab and answer technical questions. MCDV tasks will be completed through the activities of the lab, with emphasis on the differences between platforms. **Trg Resources:** • Lab Volt equipment	300

7. Test Details. EO 002.03 will be tested in EC 002.03 and PC 002 IAW table below and Annex K.

EC 002.03: Analyze Four Quadrant Control

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will complete a written assessment with emphasis on troubleshooting concepts. Questions related to circuit operation should also include aspects of troubleshooting to ensure trainees begin to adopt a “cause and effect” approach. Trainees will not have access to references for the EC.
Pass/fail criteria: 70%

ASSESSMENT RESOURCES:
- Exam

IMPACT OF FAILURE: Remedial Training / Re-test

PO 002 – Troubleshoot Analog Control Systems

EO 002.04 – Analyze Chopper Circuits

1. Performance. Analyze Chopper Circuits.
2. Conditions.
   a. Given: Stationery.
   a. Denied:
      (1) References; and
      (2) Assistance.
   b. Environmental:
      (1) Classroom;
      (2) Lab; and
      (3) MCDV alongside.
3. Standard. The trainee shall analyze chopper circuits IAW refs by:
   a. Troubleshooting buck/boost chopper control circuits;
   b. Calibrating power distribution control system (AUT0560);
   c. Maintaining MCDV Power Generation Control Systems and Components (PGCSC);
   d. Repairing MCDV PGCSC;
   e. Maintaining Power Generation Supervisory Control System (PGSCS);
   f. Maintaining CAMS; and
   g. Maintaining LSU.
4. Time Required.
   a. EO – 1210 mins to deliver training; and
   b. EC – 60 mins (total for all trainees).
5. Remarks.

   a. NFS(Q) and Sea Training have existing training material which can be used to develop lessons.

   b. Instructor shall have a detailed private tour and familiarization on an MCDV before this lesson to ensure they are comfortable with the platform and can conduct the ship's tour for the trainees.

   c. Lab experiments are used as practical formative assessments. Trainees are required to receive feedback as to their performance. This lab work is counted as the practical assessment for the EC.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Explain buck/boost chopper control theory			
1. Describe DC to DC converter theory a. Pulse Width Modulation (PWM); b. PWM power losses; 2. Identify two quadrant converter; 3. Describe four quadrant converter; a. Bi-directional operation; b. Component function; c. Verifying proper operation or failed operation through measurement; 4. Explain AC to DC converter; a. Component function; b. Filtering; c. Wave shaping – using duty cycle to control wave shape;	A50, A51, A52, C116, C117, C118, C119	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to explain the theory of Four quadrant controllers and relate theory to Naval examples (i.e. UPS system or similar). **Trg Resources:** • Nil	250
LESSON OBJECTIVE: Maintaining Buck/Boost chopper control circuits			
1. Troubleshoot buck/boost chopper control circuits 2. Calibrate power distribution control system (AUT0560) 3. Maintain PGCS and components(MCDV) 4. Adjust power generation systems and components (MCDV) 5. Repair power generation control systems and components (MCDV)	A50, A51, A52, C116, C117, C118, C119	**Method / Media:** Lab **Trg Environment:** Lab **Activity Details:** Instructor to aide trainees in the lab and answer technical questions. MCDV tasks will be completed through the activities of the lab, with emphasis on the differences between platforms. **Trg Resources:** • Lab Volt equipment	300
LESSON OBJECTIVE: Maintain PGSCS			
1. Explain what power is supplied to the ship 2. Identify the components of the PGSCS 3. Explain control method 4. Explain final resource power and how it affects the ship	A50, A51, A52, C116, C117, C118, C119	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to explain PGSCS and its components to the trainees. **Trg Resources:** • Handout/Job aid (MCDV Senior Electrician aide memoire)	200
LESSON OBJECTIVE: Maintain CAMS			
1. Explain how to trace fault signals 2. Explain what data can be received 3. Explain how CAMS relates to PGSCS	A50, A51, A52, C116, C117,	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to explain CAMS and its signal data to the trainees. **Trg Resources:** Nil	30

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
	C118, C119		
LESSON OBJECTIVE: Maintain LSU			
1. Explain how to trace fault signals 2. Explain what signals can be received 3. Explain how Local Scanning Unit (LSU) receives data from field devises and send it to CAMS 4. Explain how to operate LSU	A50, A51, A52, C116, C117, C118, C119	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to explain CAMS and its signal data to the trainees. **Trg Resources:** Nil	30
Training Activity: Ships tour onboard MCDV. Instructor will lead a tour onboard an MCDV, emphasizing differences between platforms and demonstrating MCDV related Sr Electrician procedures. Instructor should ensure all trainees are introduced to the SCR, including PLC, thyristor cabinet, and control cabinet. A senior Electrician, currently posted to an MCDV, shall be on hand to answer specific questions and help with demonstrations as required.			400

7. Test Details. EO 002.04 will be tested in EC 002.04 and PC 002 IAW table below and Annex K.

EC 002.04: Analyze chopper circuits

EC Time: 50 mins (concurrent)+ 10 min debrief

ASSESSMENT ACTIVITY: Trainees will complete a written assessment with emphasis on troubleshooting concepts. Questions related to circuit operation should also include aspects of troubleshooting to ensure trainees begin to adopt a “cause and effect” approach. At least one question on the EC should be related to the shipboard or industrial application discussed in the classroom.
Pass/fail criteria: 70%

ASSESSMENT RESOURCES: Nil

IMPACT OF FAILURE: Remedial Training / Re-test

PO 002 – Troubleshoot Analog Control Systems

EO 002.05 – Analyze Inverter Circuits

1. Performance. Analyze Inverter Circuits.

2. Conditions.

   a. Given:

      (1) Handouts;

      (2) Assistance; and

      (3) References;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and

      (2) Lab.

3. Standard. The trainee shall analyze inverter circuits IAW refs by:

   a. Explaining inverter control theory; and

   b. Maintaining control circuits to include:

      (1) Troubleshooting inverter control circuits;

      (2) Maintaining ancillary systems; and

      (3) Maintaining electrical components of systems.

4. Time Required.

   a. EO – 750 mins to deliver training

   b. EC – 60 mins (total for all trainees)

   c. PC – 910 mins (total for all trainees)

5. Remarks. Lab experiments are used as practical formative assessments. Trainees are required to receive feedback as to their performance. This lab work is counted as the practical assessment for the EC.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Explain inverter control theory			
1. Explain varying frequency for speed control; 2. Explain varying voltage with frequency (V/F) for improved speed control.	A50, A51, A52, C116, C117, C118, C119	**Method / Media:** EDI; GD **Trg Environment:** Classroom **Activity Details:** Instructor to relate inverter control theory to Naval examples (i.e. SFC system or similar) **Trg Resources:** Nil	450

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Maintain control circuits			
1. Troubleshoot inverter control circuits; 2. Maintain ancillary systems; and 3. Maintain electrical components of systems.	A50, A51, A52, C116, C117, C118, C119	**Method / Media:** Lab **Trg Environment:** Lab Volt **Activity Details:** Instructor will demonstrate relevant procedures; trainees will have an assignment to complete; the instructor will assess each trainee's lab for completeness. **Trg Resources:** • Lab volt	300

7. Test Details. EO 002.05 will be tested in EC 002.05 and PC 002 IAW table below and Annex K.

EC 002.05: Analyze inverter circuits

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: EC 002.02 will be a written assessment based on content from EO 002.02. Trainees will not have access to references or handouts for the written EC.
Trainees will be practically assessed through the lesson labs. They will have access to references and handouts during labs.
Pass/fail criteria: 70% on written; pass on all labs

ASSESSMENT RESOURCES: Nil

IMPACT OF FAILURE: Remedial Training / Re-test

PC 002: Troubleshoot analog control systems

PC Time: Part A: 100 mins + 10 mins debrief (concurrent)
Part B: 95 mins + 5 mins debrief (per trainee)

ASSESSMENT ACTIVITY:
Part A: Trainees will complete a written assessment based on the content from PO 002. References will not be provided for written assessment. Focus of the questions will be on troubleshooting aspects and equipment which are not assessed during the practical part of the PC. The written exam is to ensure trainees have a well-rounded understanding of the PO while the practical part B is to ensure the trainees are able to carry-out troubleshooting procedures.
Part B: In the Lab volt trainer, trainees will troubleshoot complex analog control circuits by calibrating analog and digital inputs and outputs for a PIC controlled RTU as used in the digital control lab. Calibration must be conducted without assistance.
Pass/fail criteria: 70% on written assessment; SAT on all checklist items for practical.

ASSESSMENT RESOURCES: Trainees will be given calibration procedure when completing Part B practical assessment.

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

MAR TECH USQ-MS Sr Elec Spec – PC 002 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Attach the umbilical-extension cable connecting the PIC trainer module to a Lab Volt power supply chassis			
Boot the RTU diagnostic module in mode 000, ensuring a starting point of mid-range (+2.5V)			
Follow procedure			
2. Ensure Switch 7, 8, and 9 are selected to be in the DC mode position to calibrate the DC mode of the analog input circuitry.			
Switch-ON the Lab Volt power supply to supply the required ±15V to the circuit card.			
Follow Procedure			
3. Ensure Switch 7, 8, and 9 are selected to be in the AC mode position as shown. This is required to calibrate the ac mode of the analog input circuitry.			
Apply +10VAC to analog input 1 (Jack#1)			
Adjust R18, the ac-trim, until pin-2 of the PIC controller zif socket = +5.0V			
4. Complete Procedure			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 – Manage unclassified networks

1. Performance. Manage unclassified networks (ALT0850).
2. Conditions.
    a. Given:
        (1) Unclassified network to include:
            (a) Node;
            (b) Server;
            (c) Switches; and
            (d) Printer.
        (2) CCTV (Closed Circuit Television) network to include:
            (a) Server;
            (b) Cabling;
            (c) Switches; and
            (d) Camera.
        (3) Test equipment; and
        (4) References.
    b. Denied: Nil.
    c. Environmental:
        (1) On ship at sea and alongside, any environmental conditions;
        (2) Shore based trainer; and
        (3) Unclassified network lab.
3. Standard. The Senior Electrical Specialist shall manage unclassified networks IAW refs by:
    a. Maintaining networks (Task# TBD) by:
        (1) Repairing control access network (ALT1020);
        (2) Maintaining network synchronization (ALT0820); and
        (3) Troubleshooting an unclassified network (ALT1930) to include:
            (a) Open Systems Interconnection (OSI) Structure; and
            (b) Transmission Control Protocol / Internet Protocol (TCP/IP).
    b. Maintaining switches (Task# TBD) by:
        (1) Operating utility software for Hirschmann switches (HT0140);
        (2) Configuring Hirschmann switches (ALT0320);
        (3) Installing Software (ALT0800); and

(4) Configuring Hirschmann switch software (ALT0350).

c. Maintaining Servers (Task# TBD) by:

(1) Repairing Servers (ALT0980); and

(2) Troubleshooting Servers (ALT1840).

d. Maintaining Nodes (Task# TBD) by:

(1) Managing Portable Operating Units (POU) (ALT0840);

(2) Configuring POU (ALT0330);

(3) Troubleshooting POU (ALT2070);

(4) Configuring printers (ALT0270);

(5) Repairing printers (ALT0940);

(6) Repairing Central Processing Unit (CPU) (ALT1010);

(7) Installing CPU (ALT0780); and

(8) Testing CPU (AUT3100).

4. References. C1, C2, C3

5. Task, Skill, Knowledge Numbers. ALT0270; ALT0320; ALT0330; ALT0780; ALT0800; ALT0820; ALT0840; ALT0850; ALT0940; ALT0980; ALT1010; ALT1020; ALT1840; ALT1930; ALT2070; AUT3100; HT0140.

6. Training Limitations.

a. CCTV cameras with switches, cabling, and server are required for training.

b. An operational network is required to conduct training and is to include:

(1) Servers,

(2) Printers,

(3) Switches,

(4) Nodes, and

(5) Various cabling.

PO 003 – Manage Unclassified Network

EO 003.01 – Maintain Network

1. Performance. Maintain a Network.
2. Conditions.
   a. Given:
      (1) Local Area Network (LAN) / Network Lab with:
         (a) Server;
         (b) Switches; and
         (c) Computers.
      (2) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Lab; and
      (2) Classroom.
3. Standard. The trainee shall maintain networks IAW refs by:
   a. Applying ethics in computer networking;
   b. Maintaining network synchronization (ALT0820);
   c. Troubleshooting a network (ALT1930) by:
      (1) Analyzing the OSI Structure; and
      (2) Investigating TCP/IP.
   d. Repairing control network access (ALT1020).
4. Time Required.
   a. EO – 1800 mins to deliver training; and
   b. EC – 60 mins (total for all trainees).

Remarks. Instructor to liaise with and send teaching points to JAG lecturer a minimum of 5 working days prior to their lecture.

5. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Apply ethics in computer networking			
2. Employ the network for its intended use a. Authorized DND use; b. Unauthorized use: i. Personal;	CAF Ethics Policy	**Method / Media:** SME discussion (JAG); GD **Trg Environment:** Classroom **Activity Details:**	60 SME 140 GD

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
ii. Games; and iii. Plugging-in USBs. c. Configuration (eg. Adding nodes); and d. Potential repercussions. 3. Describe authorized access: a. Console login; and b. User login. 4. Authorize CCTV usage: a. Use/extraction of footage; b. Deletion of footage; c. Confidentiality of videos; d. Privacy considerations of individuals captured on camera; and e. Camera control / positioning.	Defence Ethics Program CAF Ethos RCN Code of Conduct	JAG SME will discuss the legalities of computer networks including confidentiality, privacy, potential repercussions, etc. Trainees will discuss TPts, including investigating current policies by discussing validity, potential changes or additions required. **Trg Resources:** • Policy handouts	
LESSON OBJECTIVE: Maintain network synchronization (ALT0820)			
1. Define Network topologies: a. Bus; b. Star; c. Ring; d. LAN/WAN; e. Mesh; f. Peer to peer vs. client server; g. Wireless; and h. Ethernet. 2. Identify Network devices: a. Network Interface Card (NIC); b. Hub; c. Switch; d. Bridge; e. Router; f. Gateways; g. Servers; h. Workstations; and i. Peripherals.		**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to give lecture on network topologies and devices in networks including relating to ships' system devices and explaining the theory behind network synchronization. Trainees will be given worksheets with pictures of components and they will have to fill in the names. **Trg Resources:** • Component worksheets	200
LESSON OBJECTIVE: Troubleshoot a network (ALT1930)			
1. Analyze the Open System Interconnection (OSI) Structure: a. The application layer; b. The presentation layer; c. The session layer; d. The functions of the transport layer;	Network + Study Guide CompTIA+	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will instruct trainee on the OSI system. Relate the instruction to ships	400

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
e. The functions of the network layer; f. The functions of the data link layer; and g. The functions of the physical layer. 2. Analyze Point to point communication: a. Encapsulation; and b. De-encapsulation. 3. Describe Ethernet operation to include carrier sense and collision detection: a. Framing; b. Addressing; c. Cabling types; and d. Limitations of Ethernet.	Cisco CCNA Intro	systems. I.E. IPMS application vs Windows operating system. **Trg Resources:** • Cisco CCNA intro • Handout (OSI Model)	
4. Investigate TCP/IP: a. TCP/IP protocol stack; b. TCP/IP addressing; c. DNS name resolution; and d. Connection multiplexing. 5. Examine IP addressing: a. IP addressing; b. IP address components; and c. Subnetting. 6. Utilize TCP/IP utilities to determine connectivity issues: a. Ipconfig; b. Ping; c. Tracert; and d. Arp.		**Method / Media:** EDI **Trg Environment:** Classroom; LAN Lab **Activity Details:** Instructor will lecture on TCP/IP protocols to include addressing, DNS name resolution, Connection Multiplexing. The instructor will demonstrate subnetting through a Wireshark demonstration. Trainees will utilize TCP/IP utilities in the command prompt to familiarize themselves with the process. **Trg Resources:** • Cisco CCNA intro	900
LESSON OBJECTIVE: Repair control network access (ALT1020)			
1. Configuring NIC Cards; 2. Configuring licences; and 3. Solve Transmit/ Receive issues.		**Method / Media:** EDI **Trg Environment:** LAN Lab **Activity Details:** Instructor to have trainees troubleshoot issues that could arise when managing a network **Trg Resources:** • Cisco CCNA intro	100

6. Test Details. EO 003.01 will be tested in EC 003.01 and PC 003 IAW table below and Annex K.

EC 003.01: Maintain Networks

EC Time: 50 mins + 10 mins debrief (concurrent)

ASSESSMENT ACTIVITY: Assessment will be a written EC based on content from EO.003.1. Focus will be on troubleshooting techniques and processes using diagrams. References will not be available for the written assessment.
Pass/fail criteria: 60%

ASSESSMENT RESOURCES: Nil

IMPACT OF FAILURE: Remedial Training / Re-test

PO 003 – Manage Unclassified Network

EO 003.02 – Maintain Switches

1. Performance. Maintain Switches.
2. Conditions.
   a. Given:
      (1) LAN / Network Lab equipment;
      (2) Firmware for Hirshmann Switch; and
      (3) References.
   b. Denied: Nil.
   c. Environmental: Lab.
3. Standard. The trainee shall maintain switches IAW refs by operating utility software for Hirshmann Switch (HT0140) by:
   a. Enabling/ Disabling ports;
   b. Installing software (ALT0800); and
   c. Configuring Hirschmann Switches (ALT0320).
4. Time Required.
   a. EO – 100 mins to deliver training; and
   b. EC – 280 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Operate Utility Software for Hirshmann Switch (HT0140)			
1. Enabling/ Disabling ports 2. Installing software (ALT0800); and 3. Configuring Hirschmann Switches (ALT0320).	C1, C2, C3	**Method / Media:** EDI **Trg Environment:** LAN Lab **Activity Details:** Instructor will demonstrate how to complete TPts then have trainees complete the task concurrently. **Trg Resources:** • Hirschmann Switch Manual	100

7. Test Details. EO 003.02 will be tested in EC 003.02 and PC 003 IAW table below and Annex K.

EC 003.02: Maintain Switches

EC Time: 30 mins + 5 min debrief (per trainee)

ASSESSMENT ACTIVITY: Assessment will be performance based in the LAN lab. Trainees will configure switches based on EO 003.02 checklist. Trainees will complete the assessment individually (i.e. not concurrent with any other trainees) so that the invigilator can monitor their process.
Pass/fail criteria: SAT on all checklist items

ASSESSMENT RESOURCES:

- Firmware for Hirshmann switches

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 003.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Operate Utility Software for Hirshmann Switch			
Log on to utility software			
Select switch			
2. Enable/Disable ports			
Selects correct port to enable/disable			
3. Install software			
Install given software to switch			
Select switch			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 – Manage Unclassified Network

EO 003.03 – Maintain Server

1. Performance. Maintain servers (TASK# TBD).
2. Conditions.
   a. Given:
      (1) LAN / Network Lab equipment;
      (2) Functional Server;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom;
      (2) PCOT; and
      (3) Lab.
3. Standard. The trainee shall maintain servers IAW refs by:
   a. Troubleshooting servers (ALT1840) by checking connections;
   b. Repairing servers (ALT0980); and
   c. Maintaining CCTV/ Equipment Health Monitoring (EHM) Server.
4. Time Required.
   a. EO – 550 mins to deliver training; and
   b. EC – 480 mins (total for all trainees).
5. Remarks.

   a. Currently there is no CCTV server at either NFS, therefore this will have to be acquired. In the interim, instructors will show pictures of the server and explain the processes for maintenance, but the trainees will not get time to practice. Instructors could also organize a tour on an HMC Ship alongside where trainees would get to see the system in normal configuration.

   b. Currently there is no EHM database on the East coast, therefore this will have to be acquired. In the interim, the FSR will have to load historic ships' databases to the PCOT EHM server to be used for training.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Troubleshoot server (ALT0980)			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
1. Troubleshoot SQL server database on a Windows operating system a. SQL program is running b. Database is saving correctly c. Server is connected to the network	C1, C2, C3	**Method / Media:** EDI **Trg Environment:** LAN lab **Activity Details:** Instructor demonstrates methods to troubleshoot server with trainees following along. **Trg Resources:** • Handouts (SQL server How-to guide) • L3 EHM SDD publication	100
LESSON OBJECTIVE: Repair Server (ALT1840)			
1. Identify possible faults a. SQL program malfunction b. Server not talking to the network c. Temperature 2. Rectify faults	C1, C2, C3	**Method / Media:** EDI **Trg Environment:** LAN Lab **Activity Details:** Instructor to input faults in a server and have the trainees rectify. Each trainee must rectify at least one fault. **Trg Resources:** • Handouts (SQL server How-to guide) • L3 EHM SDD publication	400
LESSON OBJECTIVE: Maintain CCTV/EHM Server			
1. Manage CCTV data a. CCTV recording confirmed b. CCTV server is operational 2. Manage EHM Data a. Detach database b. Create new database c. Reattach Database d. Return to functionality	C1, C2, C3	**Method / Media:** EDI **Trg Environnent:** CCTV Server lab; PCOT **Activity Details:** Instructor will explain and demonstrate how to manage CCTV and EHM data. Trainees will carry out the processes in the Lab. **Trg Resources:** • CCTV Server • EHM Database	50

7. Test Details. EO 003.03 will be tested in EC 003.03 and PC 003 IAW table below and Annex K.

EC 003.03: Maintain Servers

EC Time: 50 mins + 10 min debrief (per trainee) **ASSESSMENT ACTIVITY**: EC 003.03 will be a performance based assessment IAW EC 003.3 checklist. In PCOT, trainees will have to manage EHM server database.

Pass/fail criteria: SAT on all checklist items

ASSESSMENT RESOURCES:

- PCOT
- EHM database

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 003.03 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Maintain EHM server			
Proper Administrator Access			
Notify proper personnel			
2. Retrieve EHM server data			
Detach database			
Create new database			
Reattach Database			
Return to functionality			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 – Manage Unclassified Network

EO 003.04 – Maintain Nodes

1. Performance. Maintain Nodes (TASK#TBD).
2. Conditions.
   a. Given:
      (1) LAN / Network Lab equipment;
      (2) Assistance; and
      (3) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) Lab.
3. Standard. The trainee shall maintain nodes IAW refs by:
   a. Managing POU (ALT0840);
   b. Configuring POU (ALT0330) by configuring start-up;
   c. Troubleshooting POU (ALT2070);
   d. Configuring printers (ALT0270);
   e. Repairing printers (ALT0940);
   f. Repairing CPU (ALT1010);
   g. Installing CPU (ALT0780); and
   h. Testing CPU (AUT3100).
4. Time Required.
   a. EO – 410 mins to deliver training; and
   b. PC – 60 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Manage portable operating units (POU) (ALT0840)			
1. Take responsibility for assets (Accountability); 2. Comply with Ship's routine; a. HR (High Readiness) Sailing b. CBRN (Chemical, Biological, Radiological, Nuclear) condition	C1, C2, C3	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor will brief ships organizations and how POU's are used under different scenarios.	130

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Bomb threat 3. Configure portable operating unit (POU) (ALT0330); a. Start-up process; and b. Restart sequence. 4. Troubleshoot portable operating unit (POU) (ALT2070): a. Hardware faults; and b. Software faults.		**Trg Resources:** Nil	
LESSON OBJECTIVE: Repair central processing unit (CPU) (ALT1010)			
1. Install central processing unit (CPU) (ALT0780) by Assembling / Disassembling; 2. Test central processing unit (CPU) (AUT3100); and 3. Reimaging CPU.	C1, C2, C3	**Method / Media:** EDI **Trg Environment:** LAN / Network lab (or classroom with stand-alone computers) **Activity Details:** Instructor to show trainees assemble/disassemble procedure and how to test a CPU. Instructor will demonstrate how to reimage a computer. Trainees will be required to reimage a computer following the steps of the instructor. **Trg Resources:** • Computer (stand-alone or network) • Computer image	250
LESSON OBJECTIVE: Repair printers (ALT0940)			
1. Changing printer cartridge 2. Configure printers (ALT0270) a. Set up IP Address; and b. Add printer / device to networked computer.	C1, C2, C3	**Method / Media:** F2F **Trg Environment:** LAN / Network lab **Activity Details:** Instructor to display how to set up a printer to a network. **Trg Resources:** • Network equipment • Networked printer	30

7. Test Details. EO 003.04 will be tested in PC 003 IAW table below and Annex K.

PC 003: Manage Unclassified Network

PC Time: 50 mins concurrent + 10 min debrief

ASSESSMENT ACTIVITY: PC 003 will be a written based assessment due to the large knowledge base of the PO. The assessment will have pictures of software that would be used and the trainee will identify the correct procedures when using the software.
Pass/fail criteria: 60%

ASSESSMENT RESOURCES: Nil

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

PO 004 – Troubleshooting Digital Control Systems

1. Performance. Troubleshooting Digital Control Systems.

2. Conditions.

   a. Given:

      (1) Operational platform;

      (2) References; and

      (3) Tools and test equipment.

   b. Denied: Nil.

   c. Environmental:

      (1) Onboard an HMC Ship;

      (2) At sea or alongside; and

      (3) Any environmental conditions.

3. Standard. The Senior Electrical Specialist shall troubleshoot Digital Control Systems IAW refs by:

   a. Programming microcontrollers by configuring software (ALT0350);

   b. Utilizing digital control for Electric Motors by repairing controllable reverse pitch propeller systems (ALT1040);

   c. Analyzing digital data (ALT0080); and

   d. Analyzing open loop digital control using packets by:

      (1) Troubleshooting propulsion control system (ALT2130);

      (2) Troubleshooting main gas turbine engine control system (AVT1440);

      (3) Repairing engine control systems (ALT1100);

      (4) Repairing transmission controls (ALT1400); and

      (5) Calibrating automatic control system (AUT0440).

4. References. C2, C.3

5. Task, Skill, Knowledge Numbers. ALT0350, ALT1040, ALT0080, ALT2130, AVT1440, ALT1100, ALT1400, AUT0440

6. Training Limitations. Nil.

PO 004 – Troubleshoot Digital Control Systems

EO 004.01 – Program microcontrollers

1. Performance. Program Microcontrollers.
2. Conditions.
   a. Given:
      (1) Microcontroller data sheets;
      (2) Assembly language data sheets; and
      (3) ASCII chart.
   b. Denied: Assistance.
   c. Environmental: Classroom.
3. Standard. The trainee shall program microcontrollers IAW microcontroller data sheets by:
   a. Configuring assembly language software (ALT0350);
   b. Comparing the functionality of PICs to microprocessors;
   c. Applying the elements of system software and memory; and
   d. Analyzing the operation of busses, oscillators and peripheral components.
4. Time Required.
   a. EO – 700 mins to deliver training; and
   b. EC – 60 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Configure assembly language Software (ALT0350)			
1. Define the three types of busses used: a. Address bus; b. Data bus; and c. Control bus: (1)R/W; (2)Clock signals; and (3)Other control signals. 2. Explain how the busses are used to synchronize information flow in a microprocessor based system; 3. Describe the registers within the microprocessor and how they interact: a. Instruction register; b. Accumulator or Working register;	C2, C3	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to cover all teaching points, asking confirmation questions. **Trg Resources:** • Handouts on busses and registers	200

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Program Counter; d. Stack pointer; e. Arithmetic Logic Unit (ALU); and f. Condition Code Register (CCR).			
LESSON OBJECTIVE: Compare the functionality of PICs to Microprocessors			
1. Comparing PICs and Microprocessors, noting similarities and differences with respect to useable memory. 2. Describe communication ports. 3. Describe input/output ports.	C2, C3	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to guide the trainees through the comparison of PIC and microprocessor block diagrams, covering off on all teaching points. **Trg Resources**: Block diagrams	100
LESSON OBJECTIVE: Apply the elements of system software and memory			
1. Describe the fetch and execute cycle used in programs. 2. Demonstrate how a PIC program would differ from a microprocessor program to load a value in a working register.	C2, C3	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor to cover all teaching points, asking confirmation questions. **Trg Resources:** Digital control system kit	50
LESSON OBJECTIVE: Analyze the operation of busses, oscillators, and peripheral components			

1. Explain how a microprocessor or PIC could be used to communicate with a device external to itself.
2. Determine the rudimentary elements of a communication protocol.
3. Describe the purpose of:
a. start bit(s);
b. stop bit(s);
c. number of data bits; and
d. error checking.
4. Explain how the 6850 Asynchronous Communication Interface Adapter (ACIA) functions.
5. Explain how the 6821 Peripheral Interface Adapter (PIA) functions.
	C2, C3	**Method / Media:** F2F
		Trg Environment: Classroom
		Activity Details: Instructor to explain and demonstrate the teaching points, asking confirmation questions. Instructor will use block diagrams of HMC Ship steering systems to link to on the job performance.
		Trg Resources:
		• Digital control system kit
		• Steering system block diagrams
			350

7. Test Details. EO 004.01 will be tested in EC 004.01 and PC 004 IAW table below and Annex K.

EC 004.01: Program microcontrollers

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: EC 004.01 will be a written assessment based on content from EO 004.01. Trainees will have access to ASCII chart, assembly language data sheets but no other references will be available.
Pass/fail criteria: 70%

ASSESSMENT RESOURCES: Stationery

IMPACT OF FAILURE: Remedial Training / Re-test

PO 004 – Troubleshooting Digital Control Systems

EO 004.02 – Utilize Digital Control for Electric Motors

1. Performance. Utilize Digital Control for Electric Motors.
2. Conditions.
   a. Given:
      (1) CCA with accessory kit;
      (2) Microcontrollers;
      (3) Microchip burner;
      (4) Cables;
      (5) RTU modules;
      (6) PIC trainer accessories;
      (7) Microcontroller data sheets;
      (8) Assembly language data sheets;
      (9) ASCII chart; and
      (10) Tool kit.
   b. Denied: Assistance.
   c. Environmental:
      (1) Classroom; and
      (2) Lab Volt.
3. Standard. The trainee shall utilize digital control for electric motors by:
   a. Examining program directives;
   b. Programming an output port;
   c. Evaluating sub-routine operation;
   d. Evaluating sub-routine operation; and
   e. Distinguishing best practices for handling interrupts.
4. Time Required.
   a. EO – 2600 mins to deliver training; and
   b. EC – 60 mins (total for all trainees).
5. Remarks.
   a. The digital control system kit includes all required materials for the labs. It will be signed for by each trainee at the beginning of the PO and signed back in after the PC.

b. Lab experiments are used as practical formative assessments. Trainees are required to receive feedback as to their performance. This lab work is counted as the practical assessment for the EC.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Examine program directives			
1. Analyzing: a. Data definition directives; b. Program structure; c. Instruction types; and d. Assembler directives.	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructor to give lecture on PIC, and demonstrate what program structure looks like, giving multiple examples and asking confirmation questions. Trainees will then be given an incomplete program and have to come up with the missing link to make it complete. **Trg Resources:** • Digital control system kit • Assembly language data sheets	750 F2F 150 Lab
LESSON OBJECTIVE: Program an output port on CCA			
1. Describe output port programming; 2. Enable LED and seven segment display driving; 3. Condition output signals; and 4. Drive larger loads.	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructor will explain and demonstrate the operation of CCA in the classroom. Trainees will operate CCA in the lab to enable LED and seven segment display. **Trg Resources:** • Digital control system kit • Assembly language data sheets • ASCII chart	300 Lecture 450 Lab
LESSON OBJECTIVE: Troubleshoot sub-routine operation			
1. Describe sub-routine functions and programming; and 2. Request sub-routines to include: a. Lookup table; and b. Corrupted memory.	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructor to cover teaching points. Trainees to conduct lab to change subroutine coding. **Trg Resources:** • Digital control system kit	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Microcontroller data sheets	
LESSON OBJECTIVE: Develop pulse width modulation control strategies			
1. Build integrated circuit, resistor ladder DACs PWM into the software. 2. Analyze resolution and linearity. 3. Describe pulse width modulation WRT CRPP.	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab Volt **Activity Details:** Instructor to cover teaching points, introducing the advantages/disadvantages of using two different motor speed control techniques. Instructor will link pulse width modulation with the CRPP and other shipboard systems as applicable. Trainees to conduct lab to build a sub-routine to PWM a motor. **Trg Resources:** • Microcontroller data sheets; • Assembly language data sheets; • ASCII chart • Digital control system kit	100 F2F 150 Lab
LESSON OBJECTIVE: Distinguish best practices for handling interrupts			
1. Explain interrupt concepts; 2. Describe PIC interrupt handling; 3. List PIC interrupt sources; 4. Demonstrate PIC interrupt configurations; 5. Analyze Timer 0, Timer 1, Timer 2; 6. Calculate Timer 0 overflow values; 7. Analyze Timer 0 with interrupts; and 8. Describe PWM setup.	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab **Activity Details:** Instructor to cover teaching points, explaining interrupts and best practices. Trainees will carry-out a lab to analyze interrupt sources and calculate timer overflow values. **Trg Resources:** • Digital control system kit • Microcontroller data sheets; • Assembly language data sheets; • ASCII chart	250 F2F 350 Lab

7. Test Details. EO 004.02 will be tested in EC 004.02 and PC 004 IAW table below and Annex K.

EC 004.02: Utilize Digital Control for Electric Motors

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: EC 004.02 will be a written assessment based on content from EO 004.02. Trainees will have access to the data sheets but will not have access to other references for the written EC.
Trainees will be practically assessed through the lesson labs. They will have access to references and handouts during labs.
Pass/fail criteria: 70% on written; pass on all lab

ASSESSMENT RESOURCES:

- Microcontroller data sheets;
- Assembly language data sheets;
- ASCII chart

IMPACT OF FAILURE: Remedial Training / Re-test

PO 004 – Troubleshooting Digital Control Systems

EO 004.03 – Analyze Digital Data

1. Performance. Analyze Digital Data (ALT0080).

2. Conditions.

   a. Given:

      (1) Microcontroller data sheets;

      (2) Assembly language data sheets;

      (3) ASCII chart; and

      (4) Digital control system kit.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) Lab.

3. Standard. The trainee shall analyze digital data IAW assembly language data sheets by:

   a. Examining the factors that establish a protocol; and

   b. Exploring the protocol used for a network.

4. Time Required.

   a. EO – 1650 mins to deliver training; and

   b. EC – 60 mins (total for all trainees).

5. Remarks. Lab experiments are used as practical formative assessments. Trainees are required to receive feedback as to their performance. This lab work is counted as the practical assessment for the EC.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Examine factors that establish a protocol			
1. Describe factors to include: a. Serial communication; b. Start, stop, parity and data bits; c. Baud rate and bps; d. RS-232 standard; e. RS 422 and 485; f. Signal levels and conversion techniques; g. Flow control techniques; h. Simplex, half-duplex and full duplex transmission; and	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab **Activity Details:** Instructor will explain and demonstrate the factors used to establish a protocol. In a lab, trainees will identify the different components. **Trg Resources:** • Digital control system kit • Assembly language data sheets	750 F2F 150 Lab

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
i. CCA module layout to include: i. Digital signal circuits (3 open drain, enhanced power MOSFETS); ii. Analog signal conditioning circuits (3 analog inputs, 2 analog outputs); iii. Onboard pulse-width-modulation capabilities; iv. Quad switch and pushbutton inputs; and v. 75C1168 serial bus transceiver chip.			
LESSON OBJECTIVE: Explore the protocol used for a network			
1. Compare single-ended versus double-ended (balanced line) for serial data transmission; 2. Develop a two-wire bidirectional bus system to include: a. Simplex mode; b. Duplex mode; c. Using Op-amp; and d. 75C1168 transceiver chip. 3. Program a packet based protocol given the two-wire bus structure; 4. Program 5-byte packet protocol; 5. Explain the 4 packet types to include: a. Transmit Data to a Specific Address (TDA); b. Request Data from a Specific Address (RDA); c. Transmit Bit Set (TBS); and d. Transmit Bit Clear (TBC) 6. Explain master-slave controller concepts; 7. Construct slave unit using the PIC based RTU modules (source code provided) 8. Describe source code framework; 9. Demonstrate packet sniffer via Visual Basic.	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab **Activity Details:** Instructor will explain and demonstrate the network protocols. In a lab, trainees will identify the network protocols and packet types including deciphering the associated code. **Trg Resources:** • Digital control system kit • Assembly language data sheets	300 F2F 450 Lab

7. Test Details. EO 004.03 will be tested in EC 004.03 and PC 004 IAW table below and Annex K.

EC 004.03 Analyze Digital Data

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: EC 004.03 will be a written assessment based on content from EO 004.03. Trainees will have access to the data sheets but will not have access to other references for the written EC.
Trainees will be practically assessed through the lesson labs. They will have access to references and handouts during labs.
Pass/fail criteria: 70% on written; pass on all lab

ASSESSMENT RESOURCES:
- Microcontroller data sheets;
- Assembly language data sheets;
- ASCII chart

IMPACT OF FAILURE: Remedial Training / Re-test

PO 004 – Troubleshooting Digital Control Systems

EO 004.04 – Analyze Open Loop Digital Control Using Packets

1. Performance. Analyze Open Loop Digital Control Using Packets.
2. Conditions.
   a. Given:
      (1) Microcontroller data sheets;
      (2) Assembly language data sheets;
      (3) ASCII chart; and
      (4) Digital control system kit.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) Lab.
3. Standard. The trainee shall analyze open loop digital control using packets by:
   a. Implementing network control of a single station; and
   b. Implementing network control of an analog device by:
      (1) Troubleshooting propulsion control system (ALT2130);
      (2) Troubleshooting main gas turbine engine control system (AVT1440);
      (3) Repairing engine control systems (ALT1100);
      (4) Repairing transmission controls (ALT1400); and
      (5) Calibrating automatic control system (AUT0440).
4. Time Required.
   a. EO – 1550 mins to deliver training;
   b. EC – 60 mins (total for all trainees); and
   c. PC – 520 mins (total for all trainees).
5. Remarks.

   a. Lab experiments are used as practical formative assessments. Trainees are required to receive feedback as to their performance. This lab work is counted as the practical assessment for the EC.

   b. Currently in the lab, there is only one simulator for a gas turbine and alternator therefore trainees must do the troubleshooting/repairing individually. Timings will change if additional simulators become available.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Implement network control of a single station			
1. Perform motor wiring and Tachometer wiring to PIC based RTU modules; 2. Poll local pushbutton switches; 3. Configure and call TBS and TBC packet transmission subroutines; 4. Control Red-Amber-Green LEDs across the network; 5. Monitor Quad Switch input values; 6. Configure and call TDA packet transmission subroutines; and 7. Configure and call RDA packet transmission subroutines.	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab **Activity Details:** Instructor will explain and demonstrate network control. In a lab, trainees will configure and supervise the network based on a given setup. **Trg Resources:** • Digital control system kit • Assembly language data sheets	250 F2F 250 Lab
LESSON OBJECTIVE: Implementing network control of an analog device			
1. Interpret tachometer reference value from reference pump (motor) using RDA packets; 2. Issue speed value to target pump (motor) using TDA packets; 3. Setup additional target pumps to illustrate redundancy capability 4. Troubleshoot propulsion control systems (ALT2130); 5. Troubleshoot main gas turbine engine control system (AVT1440); 6. Repair engine control systems (ALT1100); 7. Repair transmission controls (ALT1400); and 8. Calibrate automatic control system (AUT0440).	C2, C3	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; Lab **Activity Details:** Instructor will explain and demonstrate network control. In a lab with simulated gas turbine and alternator, trainees will configure and supervise the network based on a given setup. Trainees will each have 30 minutes to do a fault for each of the following: a. Propulsion control; b. Gas turbine; c. Engine control; and d. Transmission control. **Trg Resources:** • Digital control system kit • Assembly language data sheets	90 F2F 960 Lab (*120 per trainee*)

7. Test Details. EO 004.04 will be tested in EC 004.04 and PC 004 IAW table below and Annex K.

EC 004.04 Analyze open loop digital control using packets

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: EC 004.04 will be a written assessment based on content from EO 004.04. Trainees will have access to the data sheets but will not have access to other references for the written EC.

Trainees will be practically assessed through the lesson labs. They will have access to references and handouts during labs.
Pass/fail criteria: 70% on written; pass on all lab

ASSESSMENT RESOURCES:
- Microcontroller data sheets;
- Assembly language data sheets;
- ASCII chart

IMPACT OF FAILURE: Remedial Training / Re-test

PC 004: Troubleshooting Digital Control Systems

PC Time: 60 mins + 5 min debrief per trainee

ASSESSMENT ACTIVITY: In the lab with system simulators and IAW the references and PC checklist, the trainees shall:
a. power up the system;
b. conduct system functional checks;
c. localize 2 separate faults to lowest repairable unit;
d. repair;
e. verify operational readiness;
f. report status to assessor; and
g. power down the system.
Trainees may be required to indicate what measurements have been taken and what their plan for troubleshooting is at any time when asked by the assessor.
Assessors may ask questions related to the operation of the system at any point to assess trainees' system knowledge.
Pass/fail criteria: 70% SAT on PC Checklist items.

ASSESSMENT RESOURCES:
- Digital control system kit
- Work sheets.

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

INVIGILATOR NOTES:
1. When allocating a score, use only whole numbers. Circle the appropriate number in the score column, then transfer the value to the appropriate "Weighting" cell at the bottom of the checklist. Multiply the chosen score by the designated weight and add to the "Total" cell. Calculate the "Total Score" by adding all "Totals" together.
2. PASS / FAIL CRITERIA: Trainees must achieve a 70% SAT on checklist items to be successful with this assessment.
3. IMPACT OF FAILURE: Remedial training, re-test, TRB after second unsuccessful attempt.

PC 004 EVALUATION CHECKLIST		
Course Serial:	**Name:**	
Assessor name :	**Date :**	
Start time:	**Stop time:**	**Total time (mins):**
Equipment Fault 1:		
Equipment Fault 2:		

Initialization		
Score	**Justification**	**Remarks**
1 – 2	Did not follow correct flash up procedure. Flash up faults not recognized	
3 – 4	Followed procedure but flash up faults not recognized	
5 – 6	Followed flash up procedure. After some hesitation, corrected flash up faults	
7 – 8	Followed Flash Up procedure. Recognized and corrected flash up faults	
9 – 10	Confidently Flashed Up, efficiently corrected flash up faults.	

Use of Test Equipment		
Score	**Justification**	**Remarks**
1 – 2	Did Not select appropriate TE for Fault/ Did not demonstrate correct TE operating procedures	
3 – 4	Hesitant with selection of TE/Unsure of TE set up	
5 – 6	Selected correct TE/ Demonstrate weak operating procedures	
7 – 8	Select Correct TE/Properly utilized TE	
9 – 10	Selected Correct TE considering best options/Demonstrated thorough knowledge of operation	

Fault Finding Procedures – Fault 1		
Score	**Justification**	**Remarks**
1 – 2	Hesitant to begin Fault Finding. /Did not recognize fault indications	
3 – 4	Did not correctly interpret Fault Indications; BITE; Test Equipment Readings; Took the wrong path and did not recover.	
5 – 6	Minor inconsistencies in interpreting fault indications. May have taken wrong path but recovered. Employed logical faultfinding.	
7 – 8	Recognized and correctly interpreted fault indications. Employed Logical Faultfinding.	
9 – 10	Efficiently interpreted Fault Indications and evaluated all symptoms. Found fault with minimum number of steps.	

System Knowledge – Fault 1

3 – 4	Hesitant with selection of TE/Unsure of TE set up	
5 – 6	Selected correct TE/ Demonstrate weak operating procedures	
7 – 8	Select Correct TE/Properly utilized TE	
9 – 10	Selected Correct TE considering best options/Demonstrated thorough knowledge of operation	
Fault Finding Procedures – Fault 2		
Score	**Justification**	**Remarks**
1 – 2	Hesitant to begin Fault Finding. /Did not recognize fault indications	
3 – 4	Did not correctly interpret Fault Indications; BITE; Test Equipment Readings; Took the wrong path and did not recover.	
5 – 6	Minor inconsistencies in interpreting fault indications. May have taken wrong path but recovered. Employed logical faultfinding.	
7 – 8	Recognized and correctly interpreted fault indications. Employed Logical Faultfinding.	
9 – 10	Efficiently interpreted Fault Indications and evaluated all symptoms. Found fault with minimum number of steps.	
System Knowledge – Fault 2		
Score	**Justification**	**Remarks**
1 – 2	System knowledge did not support faultfinding. Could not answer questions from instructor. Could not properly use supporting documents	
3 – 4	System knowledge weak. Could not answer all questions. Struggled with documentation to support faultfinding. Could not fully explain fault	
5 – 6	System knowledge weak in some areas. Answered most questions from instructor and gave general explanations regarding the fault.	
7 – 8	Good system knowledge. Use of supporting documentation aided in fault finding. Explained fault clearly.	
9 – 10	Detailed system knowledge. Used documentation efficiently to find fault. Gave detailed explanation of fault	
Timing (10%)		
Score	**Justification**	**Remarks**
10	Found Fault in allotted time	
5	Found Fault with an extra 10 minutes	
Find the Fault (10%)		
Score	**Justification**	**Remarks**

10	Fault Corrected	

Score	Weighting	Total
Initialization	**X 1**	**%**
Test Equipment	**X 1**	**%**
Fault Finding – Fault 1	**X 1.5**	**%**
System Knowledge – Fault 1	**X 1.5**	**%**
Fault Finding – Fault 2	**X 1.5**	**%**
System Knowledge – Fault 2	**X 1.5**	**%**
Timing	**X 1**	**%**
Finding the Fault	**X 1**	**%**
	TOTAL SCORE	**%**

Assessor Signature:	**Trainee Signature:**

PO 005 – Manage Network Administration

1. Performance. Manage Network Administration (Task# TBD).
2. Conditions.
   a. Given:
      (1) Distributed control platform management system network;
      (2) System administrator privileges;
      (3) Human Machine Interface (HMI);
      (4) Supervision; and
      (5) References.
   b. Denied: Nil.
   c. Environmental:
      (1) On ship or ashore;
      (2) At sea or alongside; and
      (3) Any environmental conditions.
3. Standard. The Senior Electrical Specialist shall manage network administration IAW refs by:
   a. Operating IPMS applications (HT0060) by:
      (1) Managing IPMS passwords and accounts (ALT0830);
      (2) Enabling / disabling IPMS alarms (ALT0460);
      (3) Modifying IPMS alarm values (ALT0860) for:
         (a) Drills; or
         (b) Maintenance.
      (4) Setting IPMS alarm parameters (ALT1440); and
      (5) Adjusting signal conditioning coefficients (ALT0060).
   b. Operating IPMS software utilities (HT0050) by:
      (1) Operating Process Monitor (PROCMON) software (HT0100);
      (2) Operating ADT software (ALT0870); and
      (3) Configuring IPMS software (ALT0350).
   c. Analyzing IPMS EHM Data (ALT0090) by:
      (1) Retrieving and providing video evidence from CCTV networks (ALT1420);
      (2) Analyzing IPMS EHM data (ALT0090); and
      (3) Analyzing HLF data (ALT0070).

d. Operating OBTS (HT0090) by:

(1) Troubleshooting OBTS (ALT2050);

(2) Repairing OBTS (ALT1210); and

(3) Operating Orchid IS software (HT0080).

4. References. A18, C10 – C50, C83, C112.

5. Task, Skill, Knowledge Numbers. AFT0870; ALT0060; ALT0070; ALT0090; ALT0460; ALT0830; ALT0860; ALT0870; ALT1210; ALT1420; ALT1430; ALT1440; ALT2050; HT0050; HT0060; HT0080; HT0090; HT0100.

6. Training Limitations.

a. Distributed control platform management system network trainer required (PCOT).

b. Lesson objectives for PO 005, 006, and 007 must be scheduled in the order given in Ch.3 and Annex G.

PO 005 – Manage network Admin

EO 005.01 – Operate IPMS System Applications

1. Performance. Operate IPMS System Applications (HT0060).
2. Conditions.
   a. Given: IPMS enabled computer.
   b. Denied: Nil.
   c. Environmental: PCOT.
3. Standard. The trainee shall operate IPMS system applications IAW refs by:
   a. Managing IPMS passwords and accounts (ALT0830); and
   b. Modifying IPMS alarm values for drills or maintenance (ALT0860) as an administrator.
4. Time Required.
   a. EO – 60 mins to deliver training; and
   b. EC – 25 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Manage IPMS passwords and accounts (ALT0830)			
1. Navigate IPMS basic operations: a. Graphical component; b. Mimics; c. Station In Control (SIC); d. Pop up menu on mimic page; e. Role attribution; f. Messaging; g. System help and VRTU bite; h. Load analyzer; i. Synchronization status; j. Operator memo pad; and k. Record playback. 2. Create accounts as system administrator based on required user account access	C14	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will navigate through IPMS basic operations with the trainees following along. They will then create accounts based on required user account access. Each trainee will create accounts for MCC/DCC level, Watchkeeper level, System admin level. Instructor will confirm each trainee set up the accounts correctly. **Trg Resources:** • Job Aid listing steps to create an account	30
LESSON OBJECTIVE: Modify IPMS alarm values for drills or maintenance as administrator			
1. Enable / Disable IPMS Alarms (ALT0460) a. Signal override b. Signal out of service	A18, C10-C50	**Method / Media:** EDI **Trg Environment:** PCOT	30

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Inhibit alarm 2. Set IPMS alarm parameters (ALT1440) a. Warning b. High alarm c. Low alarm 3. Adjust signal conditioning coefficients based on values given from FMF engineer		**Activity Details**: Instructor will run through tpts with trainees following along on their own workstations. **Trg Resources:** Nil	

7. Test Details. EO 005.01 will be tested in EC 005.01 and PC 005 IAW table below and Annex K.

EC 005.01: Operate IPMS System Applications

EC Time: 20 mins + 5 min debrief (concurrent)

ASSESSMENT ACTIVITY: In PCOT, trainees will flash up the system and carry out a series of tasks within the system. Trainees will have to signal to assessor when they have completed a step for assessor to confirm task completion. PCs can be carried out concurrently. No references will be provided for the assessment.

Tasks to include:

- Create an administrator account
- Create new password
- Sign-in
- Change an alarm value

Pass/fail criteria: Must achieve a SAT on all checklist items

ASSESSMENT RESOURCES:

- PCOT

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 005.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Flash-up OBTS / console			
Create an administrator account			
Sign-in			
Create new password			
2. Change an alarm value			
Access correct maintenance window			
Change alarm value			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 – Manage Network Administration

EO 005.02 – Operate IPMS Software Utilities

1. Performance. Operate IPMS Software Utilities (HT0050).
2. Conditions.
   a. Given: IPMS enabled computer.
   b. Denied: Nil.
   c. Environmental: PCOT.
3. Standard. The trainee shall operate IPMS software utilities IAW refs by:
   a. Operating PROCMON software (HT0100); and
   b. Configuring IPMS software.
4. Time Required.

   EO – 140 mins to deliver training.
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Operate PROCMON software			
1. Login to PROCMON 2. Navigate PROCMON a. Locating proper utilities 3. Operate PROCMON tasks a. Select required task to perform desired function b. Allowing access to windows shell 4. Locate GMD software 5. Locate and operate IPMS Config a. Modify IPMS restart mode b. Modify debug window c. Modify shell d. Modify version e. Modify client/server f. Describe start-up process g. Describe restart sequence 6. Locate and operate ADT Software (ALT0870) a. Start/stop IPMS consoles remotely b. Start/stop PROCMON c. Modify System State d. Execute record / playback e. Execute IPMS restart mode	C10-C50	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate the navigating of PROCMON and the basic functions of the software, to include ADT function. Instructor will confirm each trainee can successfully navigate and operate PROCMON and its associated functions. **Trg Resources:** Nil	40

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
f. Modify debug window g. Execute inversion h. Modify shell i. Execute client server			
LESSON OBJECTIVE: Identify IPMS software utilities			
1. Navigate windows shell to locate: a. Cmd prompt b. ADAM.net utility c. Hyperlink d. Hirschmann e. WAGO f. PROCMON	C10-C50	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate the navigation of the windows shell to locate main IPMS software. Instructor will confirm each trainee can navigate and locate appropriate software functions on the windows shell. The lesson focus is on locating utility software only; operation will be taught in future POs. **Trg Resources:** Nil	20 (ED) 80 (I - *10 per trainee*)

7. Test Details. EO 005.02 will be tested in PC 005 IAW Annex K. There is no requirement for an EC because trainees are formatively assessed in the lesson practice.

PO 005 – Manage Network Administration

EO 005.03 – Analyze IPMS EHM Data

1. Performance. Analyze IPMS EHM Data (ALT0090).
2. Conditions.
   a. Given:
      (1) EHM server;
      (2) CCTV server; and
      (3) IPMS enabled computer network.
   b. Denied: Nil.
   c. Environmental: PCOT classroom.
3. Standard. The trainee shall analyze IPMS EHM Data IAW refs by:
   a. Retrieving data from IPMS EHM and IPMS logs (ALT1430);
   b. Analyzing EHM data (AFT0870);
   c. Analyzing HLF data (ALT0070); and
   d. Retrieving and providing video evidence from CCTV network (ALT1420).
4. Time Required.
   a. EO – 100 mins to deliver training; and
   b. EC – 320 mins (total for all trainees).
5. Remarks.
   a. Currently there is no CCTV server on both coasts and there is no EHM server on the east coast. Interim COA to mitigate this deficiency is to perform this task on a Halifax class ship alongside.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Retrieve data from IPMS EHM and IPMS logs (ALT1430)			
1. Start EHM viewer window 2. Run EHM program 3. Log into EHM server 4. Access export file utility 5. Select appropriate time frame 6. Populate with appropriate signal IDs 7. Create export file	A18, C10-C50	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate logging into the EHM server and running the EHM program. Instructor will demonstrate function of EHM data export utility. Instructor will confirm each trainee: - successfully open and log on to the EHM server, - create an export file for transfer.	40

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Trg Resources: • EHM server	
LESSON OBJECTIVE: Analyze EHM data (AFT0870)			
1. Select appropriate time frame for viewing 2. Access signal IDs and display on EHM 3. Select appropriate time frame 4. Populate with appropriate signal IDs	A18, C10-C50	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate navigation of EHM viewing window. Instructor will confirm each trainee can navigate and select appropriate signal IDs. **Trg Resources:** • EHM server	20
LESSON OBJECTIVE: Analyze HLF data			
1. Select history list search application 2. Select appropriate time frame 3. Access signal IDs and display on HLF 4. Output to event viewer 5. Select event viewer	A18, C10-C50	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate navigation of HLF viewing window. Instructor will confirm each trainee can navigate and select appropriate signal IDs **Trg Resources:** Nil	20
LESSON OBJECTIVE: Retrieve and provide video evidence from CCTV network			
1. Access IPMS CCTV export utility 2. Select appropriate time frame 3. Create export file	C14, C112	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate the IPMS CCTV file transfer utility. Instructor will confirm each trainee can create an export file for transfer. **Trg Resources:** • CCTV server	20

7. Test Details. EO 005.03 will be tested in EC 005.03 and PC 005 IAW table below and Annex K.

EC 005.03: Analyze IPMS EHM Data

EC Time: 30 mins + 10 mins debrief (per trainee) [320 mins total] **ASSESSMENT ACTIVITY**: Given a scenario in the PCOT, trainees shall log onto the EHM and retrieve appropriate signals for analysis IAW the assessment checklist. Trainees will not have access to

references or assistance during the EC. Trainees will complete the assessment individually (i.e. not concurrent with any other trainees) so that the invigilator can monitor their process.
Pass/fail criteria: SAT on all checklist items

ASSESSMENT RESOURCES:

- EHM server

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 005.03 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Log into EHM server			
Start EHM viewer window			
Run EHM program			
Log into EHM server			
2. Retrieve signals from servers			
Select appropriate time frame for viewing			
Access signal IDs and display on EHM			
3. Prepare signals for export			
Access export file utility			
Select appropriate time frame			
Populate with appropriate signal IDs			
Create export file			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 – Manage network Admin

EO 005.04 – Operate OBTS

1. Performance. Operate OBTS (HT0090).
2. Conditions.
   a. Given: IPMS enabled computer.
   b. Denied: Nil.
   c. Environmental: PCOT.
3. Standard. The trainee shall operate OBTS IAW refs by:
   a. Flashing-up OBTS in instructor mode;
   b. Troubleshooting OBTS (ALT2050);
   c. Repairing OBTS (ALT1210); and
   d. Operating Orchid IS software (HT0080).
4. Time Required.
   a. EO – 200 mins to deliver training;
   b. EC – 200 mins (total for all trainees); and
   c. PC – 40 mins (total for all trainees).
5. Remarks.

   a. A minimum of one week prior to this phase, the instructor shall confirm an FSR is scheduled to instruct the class on troubleshooting and repairing the OBTS. The instructor is to give the teaching points to the FSR so they are prepared to cover all points.

   b. For the PC: PCOT currently does not have real-time EHM capabilities therefore the trainer requires updating or the instructor / assessor can pull a pre-determined data signal from existing EHM database.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Operate OBTS (HT0090)			
1. Flash-up OBTS in instructor mode a. Log in; b. Execute Instructor mode for HMI; c. Wait for it to boot; d. Select desired HMI for student; and; e. Select IC.	A18, C14.	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate the procedure used to flash-up the OBTS software and start an HMI in student mode. Instructor shall ensure all trainees can flash up instructor mode and log onto OBTS.	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Trg Resources: Nil	
LESSON OBJECTIVE: Troubleshoot OBTS			
1. Liaise with FSR 2. Navigate help menu	C14, C28	**Method / Media:** SME lecture (FSR) **Trg Environment:** PCOT **Activity Details:** FSR will come to the PCOT to explain to the trainees how to contact them and what information to gather for expedited assistance. **Trg Resources:** Nil	50
LESSON OBJECTIVE: Repair OBTS			
1. Communicate with FSR 2. Navigate help menu	C14, C37,	**Method / Media:** SME lecturer (FSR) **Trg Environment:** PCOT **Activity Details:** Instructor will bring in an FSR to explain to the trainees on how they shall contact them and what information to gather for expedited assistance. FSR to brief on real scenarios and fixes from personal experience. **Trg Resources:** Nil	50
LESSON OBJECTIVE: Operate Orchid IS software			
1. Flash-up Orchid IS 2. Push scenario to HMI 3. Start/stop scenario	A18, C14, C106	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor will demonstrate the procedure used to flash-up the Orchid IS and push the scenario onto the student HMI. Instructor shall ensure all trainees can properly flash-up a scenario onto a student HMI. This can be demonstrated to the instructor by having the trainee start/stop the scenario on the trainees HMI. **Trg Resources:** Nil	50

7. Test Details. EO 005.04 will be tested in EC 005.04 and PC 005 IAW table below and Annex K.

EC 005.04: Operate OBTS

EC Time: 20 mins + 5 min debrief per trainee

ASSESSMENT ACTIVITY: In PCOT, from the instructor HMI, the trainee must successfully flash up the HMI in instructor mode, log into OBTS, and start the designated preloaded scenario onto the trainees' HMI. Trainees will complete the assessment individually (i.e. not concurrent with any other trainees) so that the invigilator can monitor their process.
Pass/fail criteria: SAT on all checklist items

ASSESSMENT RESOURCES:

- IPMS enabled computer

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 005.04 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Load OBTS			
Flash-up IPMS in in instructor mode			
Log onto OBTS			
Flash-up desired HMI into student mode			
2. Load Orchid IS			
Push designated preloaded scenario to student HMI			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 005: Manage Network Administration

PC Time: 30 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: In PCOT, trainees will flash up the system and carry out a series of tasks within the system. Trainees will have to signal to assessor when they have completed a step for assessor to confirm task completion. PCs can be carried out concurrently. No references will be provided for the assessment.

Tasks to include:

- Flash-up OBTS / console
- Sign-in
- Create an administrator account
- Create new password
- Shut down a computer
- Modify signal parameter
- Bring up EHM to verify modification

Pass/fail criteria: Must achieve a SAT on all checklist items

ASSESSMENT RESOURCES:

- PCOT

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

MAR TECH USQ-MS Sr Elec Spec – PC 004 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test. TRB after second unsuccessful attempt.

ITEM	SAT	UNSAT	COMMENTS
1. Flash-up OBTS / console			
Create an administrator account			
Sign-in			
Create new password			
2. Shut down a node			
Access ADT			
Select the HMI			
Shut down the correct HMI			
3. Modify signal parameter			
Select maintenance window for signal ID			
Override signal parameter			
4. Bring up EHM to verify modification / predetermined signal parameter			
Load EHM database			
Open EHM explorer window			
Select proper time and date			
Load parameter			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 006 - Troubleshoot Consoles and Devices

1. Performance. Troubleshoot Consoles and Devices.

2. Conditions.

   a. Given:

      (1) Distributed control platform management system to include:

         (a) HMI; and

         (b) LCP.

      (2) Test instruments; and

      (3) References.

   b. Denied: Nil.

   c. Environmental:

      (1) Ship at sea or alongside; or

      (2) Shore based training facilities.

3. Standard. The Senior Electrical Specialist shall troubleshoot consoles and devices IAW refs by:

   a. Repairing distributed control platform management consoles (ALT1140) by:

      (1) Assembling/disassembling HMIs (ALT0110);

      (2) Repairing HMIs (ALT1130);

      (3) Configuring the screen layout of HMIs (ALT0360); and

      (4) Repairing SKIs (ALT1360).

   b. Operating utility software for WAGO modules (HT0150) by troubleshooting WAGO modules.

4. References. A18, C16, C56, C58-C63, C65, C68, C69, C73, C74, C75, C108, C109.

5. Task, Skill, Knowledge Numbers. ALT0110; ALT0360; ALT1130; ALT1140; ALT1360; HT0150.

6. Training Limitations. Lesson objectives for PO 005, 006, and 007 must be scheduled in the order given in Ch.3 and Annex G.

PO 006 Troubleshoot Consoles and Devises

EO 006.01 - Repair IPMS consoles

1. Performance. Repair IPMS consoles (ALT1140).
2. Conditions.
   a. Given:
      (1) Stationery; and
      (2) Handouts.
   b. Denied: Nil.
   c. Environmental:
      (1) PCOT; and
      (2) IPMS Maintenance Trainer.
3. Standard. The trainee shall repair IPMS consoles IAW refs by:
   a. Repairing an HMI (ALT1130) by:
      (1) Creating and loading pre-set layouts;
      (2) Resetting the console;
      (3) Manipulating IPMS windows;
      (4) Configuring the screen layout of all HMIs (ALT0360); and
      (5) Assembling and disassembling HMIs (ALT0110).
   b. Repairing SKI (ALT1360).
4. Time Required.
   a. EO – 380 mins to deliver training; and
   b. EC – 35 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Repair HMI (ALT1130)			
1. Create and load up pre-set layouts 2. Reset console 3. Manipulate IPMS windows 4. Assemble/disassemble HMI a. Describe auxiliary components b. Configure auxiliary components c. Clean hard drive	A18, C56, C58, C59, C65, C68, C69, C73, C74,	**Method / Media:** EDI **Trg Environnent:** PCOT; IPMS Maintenance Trainer **Activity Details:** In PCOT, instructor shall describe how the HMI interface can be manipulated and repaired by the administrator and demonstrate how to create a pre-set HMI layout. Instructor	200 ED PCOT 160 I IPMS trainer (*20 per trainee*)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
	C75, C108, C109.	shall ensure trainees can manipulate HMI interface to comply with Ship standard. In the IPMS maintenance trainer, instructor shall demonstrate how auxiliary components of an HMI interact. Instructor shall allow trainees to connect and disconnect components of the HMI in the trainer to include adjusting Dip switches. **Trg Resources:** • Handouts (steps for locking/unlocking HMI windows)	
LESSON OBJECTIVE: Repair SKI (ALT1360)			
1. Describing a failed SKI 2. Replace when not working	A26, A27, C73	**Method / Media:** EDI **Trg Environnent:** PCOT **Activity Details:** Instructor shall review the operation of the SKI to include potential issues that would lead to the need for replacement. **Trg Resources:** • SKI • SKI manual for all trainees	20

7. Test Details. EO 006.01 will be tested in EC 006.01 and PC 006 IAW table below and Annex K.

EC 006.01: Repair IPMS console

EC Time: 30 mins (concurrent) + 5 min debrief

ASSESSMENT ACTIVITY: In PCOT, trainees shall create a pre-set configuration for the HMI and restart the console into the pre-set layout.
Pass/fail criteria: SAT on all checklist items

ASSESSMENT RESOURCES:
- PCOT

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 006.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
Repair IPMS consoles			
Log in as administrator			
Adjust HMI IPMS settings			
Save settings			
Restart HMI and flash up in saved pre-set configuration			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 006 - Troubleshoot Consoles and Devises

EO 006.02 - Operate Utility Software for WAGO Modules

1. Performance. Operate Utility Software for WAGO Modules (HT0150).
2. Conditions.
   a. Given: POU with software;
   b. Denied: Nil;
   c. Environmental: IPMS Maintenance Trainer.
3. Standard. The trainee shall operate utility software for WAGO modules IAW refs by:
   a. Logging onto WAGO software;
   b. Connecting to a WAGO; and
   c. Modifying WAGO inputs and outputs.
4. Time Required.
   a. EO – 260 mins to deliver training;
   b. EC – 280 mins (total for all trainees); and
   c. PC – 55 mins (total for all trainees).
5. Remarks. A WAGO is needed in the IPMS trainer for instruction. The only available WAGO at NFS(P) is in the MCRT. The manipulation of that WAGO is not ideal.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Operate Utility software for WAGO modules			
1. Describe all locations utilizing WAGOs 2. Logging into software and connecting to WAGO 3. Using software to modify input and output values	C16, C60-C63	**Method / Media:** EDI **Trg Environment:** IPMS maintenance trainer **Activity Details:** Instructor shall describe equipment that uses WAGO modules and demonstrate the use of WAGO software on WAGO modules. Trainees will operate the WAGO software on the WAGO module in the trainer with instructor supervision and feedback as required. **Trg Resources:** • Handouts (Steps for loading software)	100 ED 160 I (*20 per trainee*)

7. Test Details. EO 006.02 will be tested in EC 006.02 and PC 006 IAW table below and Annex K.

EC 006.02: Operate utility software for WAGO modules

EC Time: 30 mins + 5 min debrief per trainee (280 mins total) **ASSESSMENT ACTIVITY**: Trainee shall connect and operate the WAGO software in the IPMS Maintenance Trainer. Trainees will complete the assessment individually (i.e. not concurrent with any other trainees) so that the invigilator can monitor their process. **Pass/fail criteria**: SAT on all checklist items. **ASSESSMENT RESOURCES:** IPMS Maintenance Trainer **IMPACT OF FAILURE:** Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 006.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
Operate utility software for WAGO modules			
Shut down IPMS software			
Start-up WAGO software			
Connect to WAGO			
Manipulate WAGO inputs and outputs			
Disconnect from WAGO			
Shut down WAGO software			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 006: Troubleshoot consoles and devices

PC Time: 50 mins (concurrent) + 5 min debrief

ASSESSMENT ACTIVITY: PC 006 will be a written assessment based on content from PO 006.
Trainees will not have access to references or handouts for the written PC.
Trainees will be practically assessed during labs. During labs they will have access to references and handouts.
Pass/fail criteria: 60% on the written test; pass on all labs

ASSESSMENT RESOURCES:

- Stationery

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

PO 007 - Repair RTUs

1. Performance. Repair RTUs (ALT1350).
2. Conditions.
   a. Given:
      (1) Distributed control platform management system to include:
         (a) HMI;
         (b) RTU;
         (c) LOP; and
         (d) LCP;
      (2) Test instruments; and
      (3) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Ship at sea or alongside; or
      (2) Shore based training facilities.
3. Standard. The Senior Electrical Specialist shall repair RTUs IAW refs by:
   a. Troubleshooting RTUs (ALT2210) to include:
      (1) Operating debug tool GMD/RDT (HT0020);
      (2) Connecting / disconnecting breakout box (ALT0390); and
      (3) Assembling / disassembling RTU (ALT0120).
   b. Analyzing core software.
4. References.C9, C14, C29, C30, C31, C32, C33, C34, C35, C36, C43-C55, C57.
5. Task, Skill, Knowledge Numbers. ALT0120; ALT0390; ALT1350; ALT2210; HT0020.
6. Training Limitations. Lesson objectives for PO 005, 006, and 007 must be scheduled in the order given in Ch.3 and Annex G.

PO 007 - Repair RTU

EO 007.01 - Troubleshoot RTU

1. Performance. Troubleshoot RTU (ALT2210).
2. Conditions.
   a. Given:
      (1) PCOT;
      (2) Training RTU;
      (3) POU;
      (4) Test equipment; and
      (5) Stationery.
   b. Denied: Nil.
   c. Environmental:
      (1) PCOT; and
      (2) IPMS Maintenance Trainer.
3. Standard. The trainee shall troubleshoot RTU IAW refs by:
   a. Operating a debug tool (HT0020);
   b. Assembling and disassembling an RTU (ALT0120);
   c. Connecting and disconnecting a breakout box (ALT0390); and
   d. Finding a fault.
4. Time Required.
   a. EO – 3550 mins to deliver training;
   b. EC – 50 mins (total for all trainees); and
   c. PC – 400 mins (total for all trainees).
5. Remarks.
   a. Demonstrating how to troubleshoot a ground cannot be simulated in the lab because of the input of power, therefore the procedure and expected voltages are only instructed as theory.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Operate debug tool			
1. Load GMD software 2. Demonstrate GMD software tools and functions	C9, C14	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:**	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Instructor shall demonstrate the loading and operation of the GMD software. Instructor shall ensure the trainees can operate the GMD software. **Trg Resources:** • Handouts (Steps in debug tool use) • RTU GMD software	
LESSON OBJECTIVE: Connect/disconnect breakout box (ALT0390)			
1. Explain the breakout box 2. Connect to the breakout box 3. Utilize the breakout box	Nil	**Method / Media:** EDI **Trg Environment:** IPMS Maintenance Trainer **Activity Details:** Instructor shall demonstrate the correct use of the breakout box. Instructor shall ensure all trainees can use the breakout box **Trg Resources:** Breakout box	100
LESSON OBJECTIVE: Assemble/disassemble RTU (ALT0120)			
1. Describe RTU lay-out 2. Remove/install circuit cards	C29, C30, C31, C32, C33, C34, C35, C36, C51, C52, C53, C54, C55, C57	**Method / Media:** EDI **Trg Environment:** IPMS Maintenance Trainer **Activity Details:** Instructor shall go over the layout of the RTU, and demonstrate the removal and replacement of the circuit cards**.** **Trg Resources:** Circuit cards	100
LESSON OBJECTIVE: Troubleshoot RTU			
1. Locate Fault/alarm on IPMS 2. Use GMD/Available software 3. Produce detailed drawing of power loop associated with suspected fault 4. ID drawing and components 5. Utilize correct pins and terminal boards	C14, C56, C57	**Method / Media:** EDI **Trg Environment:** IPMS Maintenance Trainer **Activity Details:** Instructor shall input faults into the RTU trainer, and supervise and assist the trainees troubleshooting procedures. Each	50 ED 3200 I (*400 per trainee*)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
6. Verify test equipment 7. Correct pins tested in RTU 8. Fault found 9. Explain how/when to correct fault 10. Make repairs 11. Confirm RTU is operational 12. Report to supervisor 13. Follow a logical troubleshooting sequence 14. Follow all safety practices		trainee will have to troubleshoot 8 faults (they will have 50 minutes for each one). **Trg Resources:** • Training RTU; • POU; • Test equipment	

7. Test Details. EO 007.01 will be tested in EC 007.01 and PC 007 IAW table below and Annex K.

EC 007.01: Troubleshoot RTU

EC Time: 45 mins (concurrent) + 5 min debrief

ASSESSMENT ACTIVITY: EC 007.01 will be a written assessment based on content from EO 007.01. Trainees will not have access to references or handouts for the written EC.
Trainees will be practically assessed during labs. During labs they will have access to references and handouts.
Pass/fail criteria: 60% on written assessment; pass on all labs
ASSESSMENT RESOURCES: Stationery

IMPACT OF FAILURE: Remedial Training / Re-test

PC 007: Repair RTU

PC Time: 45min + 5 min debrief per trainee

ASSESSMENT ACTIVITY: In the IPMS maintenance trainer, trainees shall troubleshoot a fault in the RTU.
Pass/fail criteria: SAT on all Critical checklist items and 70% SATs on the remainder

ASSESSMENT RESOURCES:
- RTU
- Test equipment
- Networked IPMS HMI
- PPE
- Hand tools

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

MAR TECH USQ-MS Sr Elec Spec – PC 007 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 70% SATs on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test, and TRB after second unsuccessful attempt.

ITEM	SAT	UNSAT	COMMENTS
Repair RTU			
Locate Fault/alarm on IPMS			
Use GMD/Available software			
Produce detailed drawing of power loop associated with suspected fault • ID drawing and components • Correct pins and terminal boards			
Verify test equipment			
Utilize correct pins tested in RTU			
Fault found			
Explain how/when to correct fault			
Make repairs			
Confirm RTU operational			
Report to supervisor			
Follow a logical troubleshooting sequence			
*Follow all safety practices; • Correct use of meter • Use static mat • Power isolation			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:

Candidate	Name:	Date:	Signature:

PO 007 - Repair RTU

EO 007.02 - Analyze Core Software

1. Performance. Analyze Core Software.
2. Conditions.
   a. Given: Training RTU.
   b. Denied: Nil.
   c. Environmental: IPMS Maintenance Trainer.
3. Standard. The trainee shall analyze core software IAW refs by:
   a. Examining functions of core software; and
   b. Appling protocol to obtain a node ID.
4. Time Required.

   EO – 150 mins to deliver training.
5. Remarks.
   a. Ships staff are not capable of making changes to core software on ship however they are able to see the core and get information from it.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Functions of core software			
1. Access core software 2. Obtain information from the Core software 3. Apply protocol to obtain a node ID	C14, C43-C50,	**Method / Media:** EDI **Trg Environment:** IPMS Maintenance Trainer **Activity Details:** Instructor shall explain and demonstrate the functions of the Core software, to include how to access it and locate useful information. **Trg Resources:**	150

7. Test Details. EO 007.02 will be assessed through in-class activities.

PO 008 - Repair LOPs

1. Performance. Repair LOPs (ALT1160).
2. Conditions.
    a. Given:
        (1) Distributed control platform management system to include:
            (a) HMI;
            (b) RTU;
            (c) LOP; and
            (d) LCP.
        (2) Test instruments; and
        (3) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Ship at sea or alongside; or
        (2) Shore based training facilities.
3. Standard. The Senior Electrical Specialist shall repair LOPs IAW refs by:
    a. Configuring ADAM (ALT0260) by:
        (1) Testing ADAMs (ALT1450);
        (2) Programming ADAMs (ALT0920); and
        (3) Configuring ADAMs (ALT0260).
    b. Troubleshooting LOPs to include:
        (1) Consoles; and
        (2) Tablets.
4. References. C14, C60-C63, C91, C92.
5. Task, Skill, Knowledge Numbers. ALT0260; ALT0350; ALT0920; ALT1160; ALT1450; HT0130.
6. Training Limitations. Nil.

PO 008 - Repair LOP

EO 008.01 - Configure ADAM

1. Performance. Configure ADAM (ALT0260).
2. Conditions.
   a. Given: ADAMs.
   b. Denied: Nil.
   c. Environmental: IPMS Maintenance Trainer.
3. Standard. The trainee shall configure ADAM IAW refs by:
   a. Operating utility software for ADAM (HT0130) by:
      (1) Logging into utility software; and
      (2) Configuring software (ALT0350).
   b. Programing ADAM (ALT0920) by:
      (1) Following program procedure from IPMS help file;
      (2) Grounding ADAM; and
      (3) Testing ADAM (ALT1450).
4. Time Required.
   a. EO – 360 mins to deliver training; and
   b. EC – 200 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Operate utility software for ADAM			
1. Logging into Utility software 2. Selecting appropriate settings to access desired ADAM 3. Configure software (ALT0350) a. Select desired ADAM b. Change input/output	C91, C92	**Method / Media:** EDI **Trg Environment:** IPMS maintenance trainer **Activity Details:** Instructor shall demonstrate the process of loading up the ADAM software so access the desired ADAM. Instructor shall ensure trainees can load up the ADAM software so access the desired ADAM **Trg Resources:** ADAM	100 ED 160 I (*20 per trainee*)
LESSON OBJECTIVE: Program ADAM (ALT0920)			
1. Follow program procedure from IPMS help file 2. Ground ADAM	C91, C92	**Method / Media:** EDI **Trg Environment:** IPMS maintenance trainer	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
3. Test ADAM (ALT1450)		**Activity Details:** Instructor shall explain and demonstrate procedures on programing the ADAM from the IPMS help menu. Instructor shall select from the class 1-2 trainees to run though the procedure of programming ADAM. **Trg Resources:** ADAM	

7. Test Details. EO 008.01 will be tested in EC 008.01 and PC 008 IAW table below and Annex K.

EC 008.01: Configure ADAM

EC Time: 20 mins + 5 min debrief per trainee

ASSESSMENT ACTIVITY: In the IPMS Maintenance Trainer, the trainees will configure an ADAM for use. The assessor will give the ADAM type to access and the input/output to configure.
Pass/fail criteria: Must achieve all checklist items

ASSESSMENT RESOURCES:
- ADAM

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH USQ-MS Sr Elec Spec – EC 008.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
4. Configure ADAM			
Open ADAM.net utility			
Select com-port			
Change baud rate			
Select ADAM type			
Change ADAM input/output state			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 008 - Repair LOP

EO 008.02 - Troubleshoot LOP

1. Performance. Troubleshoot LOP.
2. Conditions.
   a. Given:
      (1) LOP fault diagrams; and
      (2) Stationery.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) IPMS maintenance trainer.
3. Standard. The trainee shall troubleshoot LOPs IAW refs by:
   a. Describing the components of an LOP;
   b. Drawing LOP diagrams for faults;
   c. Following a logical troubleshooting sequence;
   d. Identifying the fault in the LOP;
   e. Drawing the correct LOP fault loop; and
   f. Reporting the fault and recommendations to the chain of command.
4. Time Required.
   a. EO – 1800 mins to deliver training;
   b. EC – 50 mins (total for all trainees); and
   c. PC – 400 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Repair LOP			
1. Describe the components of the LOP a. Interaction of relays and safety disconnects b. Tablets and consoles c. Remote and local configuration 2. Describe LOP diagram for faults	C14, C60-C63	**Method / Media:** EDI **Trg Environment:** IPMS maintenance trainer; Classroom. **Activity Details:** In the trainer, the instructor shall describe the components of an LOP by pointing them out to the trainees. In the classroom, the instructor shall	150

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		show examples of LOP diagrams for LOP faults. **Trg Resources:** LOP Fault diagrams	
LESSON OBJECTIVE: Troubleshoot LOP			
1. Utilize proper documents 2. Verify test equipment 3. Identify fault 4. Draw correct LOP fault loop 5. Explain how to correct fault 6. Report to supervisor a. Explain situation (identify fault) b. Identify the impact c. Recommend solutions d. Estimate time of repair 7. Follow logical sequence a. Start at one point b. Work through procedure systematically c. Justify their process (if applicable)	C14, C60-C63	**Method / Media:** EDI **Trg Environment:** IPMS maintenance trainer **Activity Details:** Trainees shall do practical exercises consisting of potential faults that may occur. Instructor will supervise and give feedback as required. **Trg Resources: LOP fault diagrams**	50 ED 1600 I (*80 per trainee*)

7. Test Details. EO 008.02 will be tested in EC 008.02 and PC 008 IAW table below and Annex K.

EC 008.02: Troubleshoot LOP

EC Time: 45 mins concurrent + 5 min debrief

ASSESSMENT ACTIVITY: **ASSESSMENT ACTIVITY**: EC 008.02 will be a written assessment based on content from EO 008.02. Trainees will not have access to references or handouts. Trainees will be practically assessed during labs. During labs they will have access to references and handouts.
Pass/fail criteria: 60% on written assessment; Pass on all labs
ASSESSMENT RESOURCES:

- stationery

IMPACT OF FAILURE: Remedial Training / Re-test

PC 008: Repair LOP

PC Time: 45 min + 5 min debrief per trainee

ASSESSMENT ACTIVITY: Trainees shall troubleshoot a fault in the LOP.
Pass/fail criteria: SAT on all Critical checklist items and 70% SATs on the remainder

ASSESSMENT RESOURCES:

- LOP
- Maintenance Trainer
- Test equipment
- Networked IPMS HMI
- PPE
- Hand tools

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

MAR TECH USQ-MS Sr Elec Spec – PC 008 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 70% SATs on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test, and TRB after second unsuccessful attempt.

ITEM	SAT	UNSAT	COMMENTS
Repair LOP			
Locate Fault/alarm on IPMS			
Use Available software			
Produce detailed drawing of power loop associated with suspected fault • ID drawing and components • Correct pins and terminal boards			
Verify test equipment			
Correct pins tested in LOP			
Fault found			
Explain how/when to correct fault			
Make repairs			
LOP operational			
Report to supervisor			
Follow a logical troubleshooting sequence			
*Follow all safety practices; • Correct use of meter • Use static mat • Power isolation			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	
OVERALL ASSESSMENT (circle one)	**FAIL** **PASS**

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 009 - Configuring CANbus Network

1. Performance. Configuring CANbus Network (ALT0290).
2. Conditions.
    a. Given:
        (1) CANbus network to include:
            (a) PDDB;
            (b) BAMs types 1, 2, 3; and
            (c) Output devices.
        (2) Test equipment and associated software;
        (3) L3Mapps spare parts kit; and
        (4) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Ship at sea or alongside; or
        (2) Shore based training facilities.
3. Standard. The Senior Electrical Specialist shall configure a CANbus network IAW refs by:
    a. Configuring PDDBs (ALT0340) by:
        (1) Troubleshooting PDDB (ALT2080); and
        (2) Repairing PDDB (ALT1260).
    b. Configuring BAMs (ALT0280) by:
        (1) Operating Hyperlink software for BAMs (ALT0900);
        (2) Repairing BAMs (ALT0950) to include:
            (a) Configuring BAM software (ALT0350); and
            (b) Configuring BAM hardware.
        (3) Testing alarm systems to include:
            (a) IFDMS; or
            (b) BDCS.
4. References. C39, C72-C75, C80, C85, C86, C110 C39, C110.
5. Task, Skill, Knowledge Numbers. ALT0280; ALT0290; ALT0340; ALT0900; ALT0950; ALT1260; ALT2080
6. Training Limitations.
    a. No current trainer available for the IFDMS portion of training.

PO 009 - Configure CANbus Network

EO 009.01 - Configure PDDB

1. Performance. Configure PDDB (ALT0340).
2. Conditions.
   a. Given:
      (1) Handouts on PDDB; and
      (2) PDDB.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) IPMS Maintenance Trainer.
3. Standard. The trainee shall configure PDDB by:
   a. Troubleshooting PDDB (ALT2080); and
   b. Repairing PDDB (ALT1260).
4. Time Required.
   a. EO – 200 mins to deliver training; and
   b. EC – 50 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Troubleshoot PDDB			
1. Explain components of PDDB 2. Explain power distribution loop 3. Explain data distribution loop	C39, C110	**Method / Media:** F2F **Trg Environment:** Classroom **Activity Details:** Instructor shall instruct on the components and networks of the PDDB. **Trg Resources:** • Slideshow with visual aids of components • Handout on PDDB components	100
LESSON OBJECTIVE: Repair PDDB			
1. Explain installation/calibration procedure of power supply 2. Repair by replacement 3. Reset ground fault sensor	C39, C110	**Method / Media:** EDI **Trg Environment:** IPMS maintenance trainer	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Activity Details: Instructor shall demonstrate the installation and calibration of the Power supply**.** Trainees will each have to calibrate the power supply. **Trg Resources:** • PDDB	

7. Test Details. EO 009.01 will be tested in EC 009.01 and PC 009 IAW table below and Annex K.

EC 009.01: Configure PDDB

EC Time: 45 mins concurrent+ 5 min debrief

ASSESSMENT ACTIVITY: EC 009.01 will be a written assessment based on content from EO 009.01 Trainees will not have access to references or handouts. Trainees will be practically assessed during labs. During labs they will have access to references and handouts.

Pass/fail criteria: 60% on the written assessment; Pass on all labs

ASSESSMENT RESOURCES: Stationery

IMPACT OF FAILURE: Remedial Training / Re-test

PO 009 - Configure CANbus Network

EO 009.02 - Configure BAM

1. Performance. Configure BAM (ALT0280).
2. Conditions.
   a. Given:
      (1) Test equipment;
      (2) BAM network; and
      (3) POU.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) IPMS maintenance trainer.
3. Standard. The trainee shall configure BAM by:
   a. Repairing BAM (ALT0950) by configuring BAM hardware;
   b. Operating hyperlink software for BAMs (ALT0900) by configuring BAM software; and
   c. Comparing fire detection and monitoring systems on HMC Ships.
4. Time Required.
   a. EO – 2200 mins to deliver training;
   b. EC – 50 mins (total for all trainees); and
   c. PC – 440 mins (total for all trainees).
5. Remarks.
   a. Lab experiments are used as practical formative assessments. Trainees are required to receive feedback as to their performance. This lab work is counted as the practical assessment for the EC.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Repair BAM			
1. Describe BAM configuration by type within network to include LCU setup considerations 2. Explain hardware versus software faults: a. BAM lockup problems b. Software updates	C39, C72-C75, C80, C85, C86, C110	**Method / Media:** F2F **Trg Environment:** Classroom; IPMS maintenance trainer **Activity Details:** Instructor shall explain hardware and software faults, and configurations within the network. In the trainer, the instructor	200

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. BAM LED status and meaning		will demonstrate troubleshooting the faults. **Trg Resources:** • Handouts (BAM layout, procedures) • BAM • POU	
LESSON OBJECTIVE: Operate hyperlink software for BAMs			
1. Load Hyperlink on POU 2. Connect to BAM 3. Load firmware onto a BAM from a POU 4. Manipulate BAM input/outputs with hyperlink software 5. Explain utilities available on hyperlink	C39, C72-C75, C80, C85, C86, C110 C39, C110.	**Method / Media:** F2F; Lab **Trg Environment:** Classroom; IPMS maintenance trainer **Activity Details:** Instructor shall demonstrate the correct use of Hyperlink. Instructor shall ensure that all trainees can correctly connect a POU and correctly utilize Hyperlink software. Instructor shall assess the trainees' ability to troubleshoot faults on the CANbus network loop in the IPMS maintenance trainer with each trainee completing 4 BAM troubleshooting labs. **Trg Resources:** • POU with hyperlink • BAM	100 F2F 1600 Lab (*200 per trainee*)
LESSON OBJECTIVE: Compare fire detection and monitoring systems on HMC Ships			
1. Recover data from BDCS to include describing the responsibilities of controls tech and required data for incident reporting 2. Describe IFDMS setup a. Modes (local and remote) b. Alarms to fire hall	C39, C72-C75, C80, C85, C86, C110 C39, C110.	**Method / Media:** EDI **Trg Environment:** PCOT **Activity Details:** Instructor shall explain the need to recover data from BDCS for the evidence gathering team. Instructor shall demonstrate the sequence of extracting the required data for the evidence gathering team. Instructor shall ensure that all students can recover the data for the evidence gathering team. Instructor shall explain the communication sequence and control of the IFDMS, to include testing at sea and alongside. **Trg Resources:** • Pictures of IFDMS console	300

7. Test Details. EO 009.02 will be tested in EC 009.02 and PC 009 IAW table below and Annex K.

EC 009.02: Configure BAM

EC Time: 45 mins + 5 min debrief (concurrent)

ASSESSMENT ACTIVITY: EC 009.02 will be a written assessment based on content from EO 009.02 Trainees will not have access to references or handouts. Trainees will be practically assessed during labs. During labs they will have access to references and handouts.
Pass/fail criteria: 60% with the written assessment. Pass all labs.

ASSESSMENT RESOURCES:
- Stationery

IMPACT OF FAILURE: Remedial Training / Re-test

PC 009: PO Configure CANbus Network

PC Time: 50 mins + 5 min debrief (per trainee)

ASSESSMENT ACTIVITY: In the IPMS maintenance trainer, trainees shall troubleshoot a fault in the BAM loop network.

Pass/fail criteria: SAT on all Critical checklist items and 70% SATs overall

ASSESSMENT RESOURCES:
- BAM network
- Test equipment
- Hand tools

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

MAR TECH USQ-MS Sr Elec Spec – PC 009 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 70% SATs on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test, and TRB after second unsuccessful attempt.

ITEM	SAT	UNSAT	COMMENTS
1. Troubleshoot BAM			
Utilize of proper documents			
Verify test equipment			
Assess BAM condition (LEDs and physical appearance)			
Identify BAM addressing			
Flash-up Hyperlink software			
Flash-up POU			
Connect POU to BAM (Use proper sequence for HRC)			
Utilize utility software			
Identify fault			
Explain how to correct fault			
2. Report to supervisor			
Explain situation (identify fault)			
Identify the impact			
Recommend solutions			
Estimate time of repair			
3. Follow logical sequence			
Start at one point			
Work through procedure systematically			
Justify their process (if applicable)			
4. *Maintain safety practices			
• Correct use of meter • Use static mat • Power isolation			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	
OVERALL ASSESSMENT (circle one)	**FAIL** **PASS**

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

ANNEX A - *M AIN REFERENCES*

A = CAF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication	
A1	Naval Material Management System Manual Vol 1 (NAMS Policy and Procedures)	C-03-005-012/AM-001
A2	Naval Engineering Manual	C-03-005-033/AA-000
A3	Manual of Emergency Procedures Marine Systems Engineering (At sea) IPMS	C-03-050-HFX/MB-002
A4	Main Propulsion Gearing Instruments and Controls Vol 2	C-24-304-000/MS-002
A5	MODELS 226 & 246 DIFFERENTIAL PRESSURE INDICATORS	C-03-010-555/MS-001
A6	DIFFERENTIAL PRESSURE TRANSDUCER − PART NUMBER 504972 1, 504972 2, 504972 3	C-03-010-560/MS-001
A7	PRESSURE TRANSDUCER MODEL 86 PGT − AMBNC	C-03-010-561/MS-001
A8	RTD TEMPERATURE TRANSMITTER MODEL 4001PT	C-03-010-563/MS-F01
A9	TEMPERATURE TRANSMITTER MODEL SP7000RP0S24S	C-03-010-563/MS-F02
A10	INTEGRATED MACHINERY CONTROL SYSTEM RELAY BOXES	C-03-010-566/MS-F01
A11	DIFFERENTIAL PRESSURE TRANSDUCER	C-03-010-567/MS-F01
A12	GEMS TANK LEVEL INDICATING SYSTEMS 36000 SERIES TLI SYSTEMS	C-03-010-571/MS-F01
A13	FLOW SWITCHES AND FLOW CONTROLS	C-03-010-572/MS-F01
A14	LIQUID LEVEL SWITCHES FOR MARINE SERVICE GEMS	C-03-010-573/MS-F01
A15	FLUID SYSTEM PRESSURE TRANSDUCERS	C-03-010-585/MS-001
A16	IPMS Relay Boxes	C-03-010-566/ MS-F01
A17	DIFFERENTIAL PRESSURE TRANSDUCER	C-03-010-575/MS-F01
A18	Drill Manual	C-03-050-HFX/MV-002
A19	Main Propulsion System Manual HFX Class	C-24-293-000/MS-001
A20	Main Gearing SIGNAL CONDITIONING UNIT Preventative Maintenance Performance Test	C-24-301-A00/NK-001
A21	Main Propulsion Gearing Manual Vol 1	C-24-302-000/MS-001
A22	Main Propulsion Gearing Manual Vol 2	C-24-302-000/MS-002

A23	Main Gearing Signal Conditioning Unit Model Scu-020	C-24-302-A00/MS-001
A24	Main Propulsion Gearing Instruments and Controls Vol 1	C-24-304-000/MS-001
A25	Operation and Maintenance Instructions with Parts Identification List Controllable Pitch Propellers and Propeller Shafting	C-24-559-000/MS-001
A26	BRIDGE AND COMBAT CENTRE ORDER AND ACTUAL KNOTS INDICATOR	C-27-A14-000/MS-001
A27	Bridge and Combat Center Order and Actual Knots Indicator	C-27-A14-000/MS-001
A28	Steering Gear Control System, Book 1	C-27-778-000/MS-001
A29	Steering Gear Control System, Book 6	C-27-778-000/MS-001
A30	Main Gearing Signal Conditioning Unit	C-23-302-A00/MS-001
A31	Power supply assembly, uninterruptible power supply 1kw AC/DC part number 857-117-001 – applicable to Halifax Class	C-38-267-000/MS-000
A32	Recovery, Assist, Securing and Traversing (RAST) System manual	C-39-158-000/MS-001
A33	1KW UPS Over current protection card	C-38-267-000/MS-000
A34	Degaussing Power Supply schematics	C-38-232-000/MS-002
A35	Helo Hauldown tension amplifier circuit	C-39-158-000/MS-001
A44	MARCORDs	
A45	Naval Material Management System (NaMMS)	C-03-005-012/AM-001
A50	THYRISTOR CONVERTERS – MANUAL CUBICLES SCR-1 AND SCR-2	C-25-569-000/MF-005
A51	OPERATION AND MAINTENANCE MANUAL – FREQUENCY CONVERTER P NUMBER 1003720300 – TRANSFORMER/CONTACTOR ASSEMBLY PART NUMBER 1003720210 (T/CA)PN 1003790210 APPLICABLE TO HALIFAX CLASS	C-38-288-000/MS-001
A52	1KW DC Uninterruptible Power Supply	C-38-299-000/MS-000
C1	Deal Richard, Cisco Certified Network Associate, McGraw Hill	978-0071497305
C2	Bateson Robert, Introduction to Control System Technology, 3rd Edition	978-0130306886
C3	Introductory Electronic Devices and Circuits, 6th Edition, (Paynter, Robert	978-0130617507
C4	Basic Technical Mathematics with Calculus, 8th Edition (Washington, Allyn J.)	978-0321306890
C5	Network+ Study Guide 3rd Ed (Goth, David)	978-0782140149
C6	Windows XP OS Manual	
C7	LCU Calibration Procedure	CD268-102624-01
C8	Hardware Design Document	CD300-100793-01
C9	Interface Control Document	CD300-100797-01
C10	HCI Software Design Document (Volume 1) for HMI	CD300-100799-01

C11	Software Design Document for the Control Application Software (CAPS)	CD300-100800-01
C12	Software Design Document (Volume 1) for the Equipment Health Monitoring (EHM)	CD300-100811-01
C13	Software Design Document for the Plant Systems Machinery Simulation	CD300-100812-01
C14	Software User's Manual (SUM)	CD300-100812-01
C15	Software Design Document of The Serial Link Interfaces	CD300-101169-01
C16	Software Design Document (Volume 19) for the WAGO Modbus (WAGO)	CD300-101177-01
C17	Software Design Document (Volume 6) for the Controllable Reversible Pitch Propeller Controller (CRPPC) CSC of the Control Application Software (CAPS) CSCI	CD300-101194-01
C18	Software Design Document (Volume 4) for the Power Lever Actuator Controller (PLAC) CSC of the Control Application Software (CAPS) CSCI	CD300-101195-01
C19	Software Design Document (Volume 2) for the Display Manager (Dm)	CD300-101202-01
C20	Software Design Document (Volume 3) for the Machinery Control Manager (MCM)	CD300-101203-01
C21	Software Design Document (Volume 4) for the Damage Control Manager (DCM)	CD300-101204-01
C22	Software Design Document (Volume 5) for the Data Logger (Dl)	CD300-101205-01
C23	Software Design Document (Volume 6) for the Trending (TRD)	CD300-101206-01
C24	Software Design Document (Volume 7) for the Event Manager (EVM)	CD300-101207-01
C25	Software Design Document (Volume 8) for the Event Viewer (EVV)	CD300-101208-01
C26	Software Design Document (Volume 9) for the HMI Common Data Manager (HCDM)	CD300-101209-01
C27	Software Design Document (Volume 10) for the Remote View and Record and Playback (RVRNP)	CD300-101210-01
C28	Trainer Design Document (TDD)	CD300-101220-01
C29	Software Design Document (Volume 2) for the Equipment Health Monitoring (EHM) CSCI	CD300-101227-01
C30	Software Design Document (Volume 8) for the Electrical Power Control System (EPCS) CSC of the Control Application Software (CAPS) CSCI	CD300-101234-01
C31	Software Design Document (Volume 2) for the Propulsion System Coordinator (PSC) CSC of the Control Application Software (CAPS) CSCI	CD300-101238-01

C32	Software Design Document (Volume 3) for the Engine Sequencer (ESEQ) CSC of the Control Application Software (CAPS) CSCI	CD300-101239-01
C33	Software Design Document (Vol. 5) for the Vibration Monitoring Function (Vibmon) CSC of the Control Application Software (Caps)	CD300-101240-01
C34	Software Design Document (Volume 7) for the Diesel Generator Local Control Panel (DGLCP) CSC of the Control Application Software (Caps)	CD300-101247-01
C35	Software Design Document (Volume 10) for the Damage Control System of the Control Application Software (CAPS) CSCI	CD300-101254-01
C36	Software Design Document (Volume 9) for the Automatic Sequence Controller (ASC) CSC of the Control Application Software (CAPS)	CD300-101255-01
C37	Trainer User Documentation (TUD) Halifax	CD300-101285-01
C38	IPMS Interconnect Document	CD300-101305-01
C39	DCS Interconnect Document	CD300-101305-02
C40	O&I, LSOCP	CD300-101775-01
C41	O&I, HFCP	CD300-101776-01
C42	Recommended Tools and Test Equipment List	CD300-102031-01
C43	Core Design Document Vol1	CD389801.8.300
C44	Core SDD Vol. 2 For Built in Test CSC	CD389802
C45	Core SDD Vol.3 For Initialization CSC	CD389803
C46	Core SDD Vol.4 For Communication Handler CSC	CD389804
C47	Core SDD Vol.5 For Database Handler CSC	CD389805
C48	Core SDD Vol.6 For Input/output Handler CSC	CD389806
C49	Core SDD Vol.8 For Application Program Interface CSC	CD389808
C50	Core SDD Vol.9 For Support CSC	CD389809
C51	CCA (VSVI)	CD-CD299028018268-1--
C52	CCA (VAI-48)	CD-CD299030018268-1-B
C53	CCA (VDI-48)	CD-CD299034018268-1-A
C54	CCA (VDOP-24)	CD-CD299036018268-1-A
C55	CCA (VLLI-24)	CD-CD299038018268-1-F
C56	Halifax Class IPMS Interconnect Schematic Diagram	ED-899-100650-01-1-C
C57	Halifax Class RTU Schematic Diagram	ED-899-100650-02-1-A
C58	Supervisor's Console Schematic Diagram	ED-899-100650-05-1-A
C59	Machinery Control Console Schematic Diagram	ED-899-100650-06-1-B
C60	Quartermaster LOP Schematic Diagram	ED-899-100650-09-1-B

C61	Schematic Diagram AER LOP	ED-899-100650-10
C62	Halifax Class AER LOP Schematic Diagram	ED-899-100650-10-1
C63	Halifax Class FER LOP Schematic Diagram	ED-899-100650-11-1-A
C64	Switch Enclosure Schematic Diagram	ED-899-100650-13-1
C65	Halifax Class BCC Schematic Diagram	ED-899-100650-14-1
C66	LSOCP Electrical Schematic	ED-899-100650-15
C67	LSOCP Schematic Diagram	ED-899-100650-15-1
C68	HFCP Electrical Schematic	ED-899-100650-16
C69	HFCP Schematic Diagram	ED-899-100650-16-1-A
C70	Shock Isolated Cabinet Schematic	ED-899-100650-21-1
C71	Large Screen Display Schematic	ED-899-100650-22-1
C72	Damage Control System Schematic Diagram	ED-899-100550-12-1
C73	Halifax Class IPMS Interconnect Schematic Diagram	ED-899-100650-01-1-D
C74	Damage Control Console Schematic Diagram	ED-899-100650-03-1-B
C75	Damage Control HQ2 Console Schematic Diagram	ED-899-100650-07-1-B
C76	DG LCP Schematic Diagram	ED-899-100650-08-1-C
C77	Electrical Switchboard Controller (ESC) Schematic Diagram	ED-899-100650-17-1-B
C78	DSC Power Distribution Schematic Diagram	ED-899-100650-18-1-A
C79	Kit, AC Ups & Converter	L4491-MA330-101788-01
C80	Kit, BAM Installation	MA330-100605-01
C81	LSOCP Electrical Installation	MA330-101735-01
C82	HFCP Electrical Installation	MA330-101736-01
C83	Orchid IS User Manual Version 5.10.0 Rev L. PDF	N/A
C84	VME Pentium-M Single Board Computer	PS268-100169-01
C85	BUS Access Module Type 1	PS268-101593-01
C86	BUS Access Module Type 2	PS268-101593-02
C87	Module, 2-Channel Input Module	PS338730085804
C88	Rugged USB 61-Key Panel Mount Keyboard	PS520-100626-01
C89	Rugged Computer, 4U Rackmount	PS804-100625-01
C90	Rugged Rackmount Server	PS804-100796-01
C91	Rugged Portable Computer	PS804-101032-01
C92	Rugged Tablet Computer	PS804-101128-01
C93	Rugged Integrated Computer/ Large Format LCD	PS804-101223-01
C94	Shock Isolated Equipment Cabinet Assembly	PS815-101282-01
C95	PSCD, LSOCP Cabinet	PS815-101749-01
C96	PSCD, HFCP Cabinet	PS815-101754-01
C97	Managed Ethernet Switch DIN Rail Mount	PS842-100824-04-1
C98	Rugged Compact POE IR Camera	PS864-101764-01
C99	POE Thermal Camera	PS864-102471-01
C100	Rugged Compact POE Thermal Camera	PS864-102471-01
C101	Module ,8 Channel Digital Input	PS-PS338730165804-1
C102	Module, 2-Channel Analog IP 4-20MA	PS-PS338730165804-1

C103	Data Acquisition DIN Rail Module	PS-PS613-100670-01-1
C104	Ethernet TCP/IP Fieldbus Coupler	PS-PS804-100702-01
C105	Ethernet Switch	PS-PS842-100824-01
C106	Orchid SE User Manual	SD201179L01
C107	Console Unit Test Procedure	TS815-102621-01
C108	Diesel Generator Local Control Panel (DG LCP) Test Procedure	TS815-102622-01
C109	Quarter Master Local Control Panel (QM LOP) Unit Test Procedure	TS815-102622-02
C110	Power Data Distribution Box (PDDB) Unit Test Procedure	TS815-102623-01
C111	Switch Enclosure Test Procedure	TS815-103324-01
C112	CCTV User Manual	W8483-076370/B
C113	L3 Maps IPMS Operation & Maintenance Manual	
C114	Statement of Work Enhancements to IPMS Maintenance Trainer	
C115	Microchip PIC 16F87XA Datasheet, Microchip Technology inc, 2003	
C116	Paynter, Robert T., Introductory Electronic Devices and Circuits, 6th Edition, Pre Hall, 2003	
C117	Tocci, Ronald J, Digital Systems, Principles and Applications, 5th Edition, Prent Hall, 1990	
C118	Wildi, Electrical Machines, Drives and Power Systems, 6th Ed, Pearson Education, Inc., 2006 (ISBN 0-13-177691-6)	
C119	Washington, Allyn J., Basic Technical Mathematics with Calculus, 8th Edition, Pearson, 2005	
D1	Detailed Training Needs Analysis (DTNA) Report for the Halifax Class	
	Integrated Platform Management System (IPMS)	
D2	Fleet Service Report (FSR Notes)	
D3	CFNES Standard Calibration Procedure	

ANNEX B - *TASK LIST*

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
AFT0870	Analyze equipment health monitoring data	T		005
AKT0330	Calibrate temperature sensors	T		002
ALT0060	Adjust signal conditioning coefficients	T		005
ALT0070	Analyze HLF Data	T		005
ALT0080	Analyze digital data	T		004
ALT0090	Analyze IPMS EHM data	T		005
ALT0110	Assemble / disassemble human-machine Interface (HMI)	T		006
ALT0120	Assemble / disassemble remote terminal unit (RTU)	T		007
ALT0140	Calibrate control sensors	T		002
ALT0260	Configure analog digital acquisition modules (ADAM)	T		008
ALT0270	Configure ~~B/W and colour~~ printers	T	Remove qualifier = redundant	003
ALT0280	Configure bus access modules (BAMS)	T		009
ALT0290	Configure CANbus network	T		009
ALT0320	Configure Hirschman switches	T		003
ALT0330	Configure portable operating unit (POU)	T		003
ALT0340	Configure power data distribution boxes (PDDB)	T		009
ALT0350	Configure software	T		008
ALT0360	Configure the screen layout of all human machine Interfaces	T		006

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
ALT0390	Connect/disconnect breakout boxes	T		007
ALT0460	Enable/disable IPMS alarms	T		005
ALT0780	Install central processing unit (CPU)	T		003
ALT0800	Install software	T		003
ALT0820	Maintain network synchronization	T		003
ALT0830	Manage IPMS passwords and accounts	T		005
ALT0840	Manage portable operating units (POU)	T		003
ALT0850	Manage unclassified network	T		003
ALT0860	Modify IPMS alarm values, for drills or maintenance	T		005
ALT0870	Operate ADT software	T		005
ALT0900	Operate hyperlink software for BAM's	T		009
ALT0920	Program analog digital acquisition modules (ADAM)	T		008
ALT0940	Repair ~~B/W and colour~~ printers	T	Remove qualifier = redundant	003
ALT0950	Repair bus access modules (BAMS)	T		009
ALT0980	Repair ~~CCTV~~ Server	T	Remove qualifier = valid for all servers	003
ALT1010	Repair central processing unit (CPU)	T		003
ALT1020	Repair control access network	T		003
ALT1100	Repair engine control systems	T		004
ALT1130	Repair human-machine Interface (HMI)	T		006
ALT1140	Repair integrated platform management system (IPMS) console	T		006
ALT1160	Repair LOP (Fwd/Aft engine room)	T		008

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
ALT1210	Repair Onboard training system (OBTS)	T		005
ALT1260	Repair power data distribution Boxes (PDDB)	T		009
ALT1350	Repair remote terminal unit (RTU)	T		007
ALT1360	Repair shaft knot indicators	T		006
ALT1400	Repair transmission controls	T		004
ALT1420	Retrieve and provide video evidence from Closed circuit Television (CCTV) network	T		005
ALT1430	Retrieve data from IPMS equipment health monitoring (EHM) and IPMS logs	T		005
ALT1440	Set IPMS alarm Parameters	T		005
ALT1450	Test analog digital acquisition modules (ADAM)	T		008
ALT1840	Troubleshoot ~~CCTV~~ Server	T	Remove qualifier = valid for all servers	003
ALT1930	Troubleshoot ~~fibre optic~~ network	T	Remove qualifier = valid for all networks	003
ALT2050	Troubleshoot onboard training system (OBTS)	T		005
ALT2070	Troubleshoot portable operating unit (POU)	T		003
ALT2080	Troubleshoot power data distribution boxes (PDDB)	T		009
ALT2130	Troubleshoot propulsion control system	T		002, 004
ALT2210	Troubleshoot remote terminal unit (RTU)	T		007
AUT0440	Calibrate automatic control system	T		004
AUT0470	Calibrate circuit cards	T		002
AUT0560	Calibrate power distribution control system	T		002

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
AUT0580	Calibrate power generation control system	T		002
AUT3100	Test central processing unit (CPU)	T		003
AVT1440	Troubleshoot main gas turbine engine control system	T		004
HT0020	Operate debug tool GMD/RDT	T		007
HT0050	Operate integrated platform management systems (IPMS) software utilities	T		005
HT0060	Operate integrated platform management systems (IPMS) systems applications	T		005
HT0080	Operate Orchid Instructor Station (IS) Software	T		005
HT0090	Operate OBTS	T		005
HT0100	Operate process monitoring Software (PROCMON)	T		005
HT0130	Operate utility software for analog digital acquisition modules (ADAM)	T		008
HT0140	Operate Utility Software for Hirshmann Switch	T		003
HT0150	Operate Utility Software for WAGO modules	T		006
TBD	Troubleshoot Analog Control Systems	T	To be added to JTAR, main task for Sr Electrician	002
TBD	Perform Power Calculation	T	To be added to JTAR, main task for Sr Electrician	002
TBD	Perform trials on SCR propulsion cabinet	T	To be added to JTAR, main task for MCDV Sr Electrician	002
TBD	Maintain power distribution system	T	To be added to JTAR, main task for Sr Electrician	002

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
TBD	Maintain electrical propulsion control systems	T	To be added to JTAR, main task for Sr Electrician	002
TBD	Maintain the propulsion system	T	To be added to JTAR, main task for Sr Electrician	002
TBD	Maintain power generation control systems and components	T	To be added to JTAR, main task for Sr Electrician	002
TBD	Maintain Propulsion control systems and components	T	To be added to JTAR, main task for Sr Electrician	002
TBD	Maintain networks	T	To be added to JTAR, main task for Sr Electrician	003
TBD	Maintain switches	T	To be added to JTAR, main task for Sr Electrician	003
TBD	Maintain Servers	T	To be added to JTAR, main task for Sr Electrician	003
TBD	Maintain Nodes	T	To be added to JTAR, main task for Sr Electrician	003
TBD	Manage Network Administration	T	To be added to JTAR, main task for Sr Electrician	005
TBD	Maintain PGSCS	T	To be added to JTAR, main task for Sr Electrician	005
TBD	Troubleshoot LOPs	T	To be added to JTAR, main task for Sr Electrician	008
AFT0330	Prepare pre-refit survey reports (650s)	NT	Trained at RQ-PO2	
AFT1240	Draft engineering change proposal	NT	Not performed by Sr Elec	
AFT1270	Draft operational deficiency report (OpDef)	NT	Previously trained (AB)	
AFT1280	Draft pre-installation failure (PIF) reports	NT	Previously trained (LS)	
AFT1310	Draft technical documentation	NT	Previously trained (AB)	

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
AKT1070	Install PLC program	NT	No PLC programs to install	
ALT0010	Adjust central alarm monitoring system printers	NT	Done by ISSC	
ALT0020	Adjust control sensors	NT	Previously trained (LS)	
ALT0030	Adjust control sensors mechanical components	NT	Previously trained (LS)	
ALT0050	Adjust power lever actuators	NT	Previously trained (LS)	
ALT0100	Assemble / disassemble gearbox pneumatic control system components	NT	Previously trained (LS)	
ALT0130	Calibrate analog measuring test equipment	NT	Done by TEMMES	
ALT0150	Calibrate controllable reversible pitch propeller control system components	NT	Done by FMF	
ALT0160	Calibrate engine control systems	NT	Done by FMF	
ALT0170	Calibrate gearbox pneumatic control system	NT	Done by FMF	
ALT0180	Calibrate integrated platform management system monitors	NT	Done by FMF	
ALT0190	Calibrate PDE control components	NT	Done by FMF	
ALT0200	Calibrate signal conditioners	NT	Done by FMF	
ALT0250	Complete integrated platform management systems (IPMS) pre-sailing checks	NT	Checklist is in SSOs (redundant)	
ALT0300	Configure CCTV network and equipment	NT	Previously trained (LS)	
ALT0310	Configure fibre optic network	NT	Lower level	
ALT0370	Configure WAGO modules	NT	Previously trained (LS)	
ALT0520	Inspect central alarm monitoring system	NT	Checklist / job aid available	
ALT0930	Repair analog digital acquisition modules (ADAM)	NT	Done by FSR	

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
ALT0960	Repair Cat 5e cable network	NT	Previously trained (LS)	
ALT0970	Repair CCTV network and equipment	NT	Redundant ALT1430	
ALT0990	Repair central alarm monitoring system	NT	Done by ISSC	
ALT1000	Repair central alarm monitoring system Printers	NT	Done by ISSC	
ALT1030	Repair control transformer	NT	Previously trained (LS)	
ALT1040	Repair controllable reversible pitch propeller systems (IPMS)	NT	Previously trained (AB)	
ALT1060	Repair DG local control panel	NT	Previously trained (LS)	
ALT1080	Repair digital control system	NT	Previously trained (LS)	
ALT1110	Repair fibre optic network	NT	Lower level	
ALT1120	Repair helo fuel/de-fuel control panel (HFCP)	NT	Previously trained (LS)	
ALT1150	Repair integrated platform management system (IPMS) monitor	NT	Redundant ALT1130	
ALT1170	Repair MOB system	NT	Lower level	
ALT1180	Repair momentum modules	NT	Task is to replace, not repair	
ALT1190	Repair motherboard	NT	Task is not performed	
ALT1200	Repair network communication cables	NT	Previously trained (LS)	
ALT1220	Repair Onboard training system (OBTS) Software	NT	Lower level	
ALT1230	Repair pin connectors	NT	Lower level	
ALT1240	Repair pneumatic control components	NT	Previously trained (LS)	
ALT1250	Repair portable operating unit (POU)	NT	Replace, not Repair	
ALT1270	Repair power lever angle actuators	NT	Trained on LM2500 maintenance course	

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
ALT1280	Repair precision control equipment	NT	Previously trained (LS)	
ALT1330	Repair propulsion diesel engine starter box	NT	Previously trained (LS)	
ALT1340	Repair QM local operating panel	NT	Redundant	
ALT1370	Repair shore based trainer	NT	Repaired by L3	
ALT1410	Restart IPMS equipment after power failure	NT	Automatic function	
ALT1510	Test central alarm monitoring system	NT	Done by ISSC	
ALT1520	Test components with built-in test equipment (BITE)	NT	Automatic function	
ALT1600	Test human-machine Interface (HMI)	NT	Nothing to test on HMI	
ALT2040	Troubleshoot momentum modules	NT	Previously trained (LS)	
ALT2060	Troubleshoot operating irregularities of the propulsion system (IPMS)	NT	Previously trained (LS)	
ALT2090	Troubleshoot power lever angle actuators	NT	Trained on LM2500 maintenance course	
ALT2100	Troubleshoot precision control equipment	NT	Previously trained (LS)	
ALT2120	Troubleshoot programmable logic controller	NT	Previously trained (LS)	
ALT2170	Troubleshoot propulsion diesel engine enclosure carbon monoxide sensor	NT	Previously trained (LS)	
ALT1090	Repair electrical switchboard control (ESC)	NT	Not performed by Sr Elec	
AUT1140	Inspect final resource system charger (MCDV PLC ups battery charger)	NT	Done by ISSC	
AUT3210	Test digital Input/output processor (WAGO)	NT	Previously trained (LS)	
AUT3350	Test final resource system charger (MCDV PLC ups battery charger)	NT	Done by ISSC	

Task No	Task Statement	Train / No Train	NT Reasoning / Comments	PO
AVT0460	Calibrate main propulsion gas turbine components	NT	Not performed by Sr Elec	
HT0110	Operate Scenario manager program	NT	Not performed by Sr Elec	
HT0120	Operate shore based trainer	NT	Experiential	
HT0160	Operate Vibration Monitor Suite	NT	Not performed on the ship	
OT0210	Supervise liquid cargo transfer (SEMS)	NT	Performed by watchkeeper	

ANNEX C - *QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS*

Dates: 16 May – 1 Jun 18

Location: CFB Esquimalt, Workpoint, Haida Building, NTDC(P) MS Div

Board Members:

Position	Rank and name	MOSID / Unit	Dates
Occupation Manager	CPO1 Byron Pryor	00379 / D Nav P&T	28 May – 1 Jun
Training Manager	CPO2 Yannick Gagné	00379 / NPTG HQ	Not present; available by email
Chair	CPO2 James Zsombor	00379 / NTDC(P)	16 May – 1 Jun
SME	PO1 Ken Penney	00379 / NTDC(P)	16-23, 28 May – 1 Jun
SME	PO2 Curt Gillies	00379 / NTDC(P)	14 May – 1 Jun
SME	PO2 Marty Magnan	00379 / NTDC(P)	22, 24 May – 1 Jun
SME	MS Kevin Solski	00379 / FMF Cape Breton	14 May – 1 Jun
SME	MS Shawn Malone	00379 / NFS(P)	22 – 30 May
TDO	SLt Sarah Valentine	00211 / NTDC(P)	14 May – 1 Jun
Secretary	SLt Kyle Knight	00211 / NTDC(P)	16, 22 – 25 May

Board Objective:

The aim of this board is to analyze the job performance expected of Senior Electricians onboard HMC Ships and design complementary training to enable each performance.

Discussions / Decisions:

For the purposes of this RoP, the following PO nomenclature applies:

PO 001 “Interpret Circuit Fundamentals”
PO 002 “Troubleshoot analog control systems”
PO 003 “Manage unclassified networks”
PO 004 “Troubleshoot digital control systems”
PO 005 “Mange network administration”
PO 006 “Troubleshoot consoles and devices”
PO 007 “Repair RTUs”
PO 008 “Repair LOPs”
PO 009 “Configure CANbus networks”

**Decisions or discussions which are stricken through were changed at a later date*

Decision/Event	Discussion / Detail
10 May 18	
MCDV material	Discussion as to whether or not the MCDV OJT should remain as is or be shortened / changed. Reviewed the email in the RoP of the legacy QL6A from then PO1 Perron which stated that an electrician posted to an MCDV required more experience on the following topics: a. PLC/PIC b. Electrical motors c. Field Devices d. Circuit cards / control systems e. Drills

Decision/Event	Discussion / Detail
	After further discussion with CPO2 Perron (currently posted NTDC(P)), it was decided that PLC/PIC training would be fixed by a job aid which stated the new vocabulary for Kingston Class; Electrical motors and Field devices are now taught at the LS level; Circuit cards are covered by the IPMS portion of training; and Drills are initiated the same on all platforms (on MCDV, Sr Elec only initiates drills) and steps are trained throughout the troubleshooting lessons.
	Recommendation which went forward to OA and Trg Manager (with support from NTDC(P) MS Div Cdr) was to remove the OJT portion of training and ensure ~~ISSC contracts, WEAF theory~~ *[*Note: Board later decided ISSC contracts and WEAF theory were included in previous training]*, and possibly a tour was included in the training.
Calibrate	For this QSP, "Calibrate" is a term for use in a lab. On the job, the MAR TECH MS Elec will not be calibrating the equipment as it is done by the field services representative, however they need to know how to calibrate the trainer so that it works correctly and so that they are better able to adjust components of a system when on ship.
14 May 18	
ALT1930	"Troubleshoot fibre optic network" Was NT on the JTAR but is T because it is at a different level than LS since they will be using the network to troubleshoot instead of a meter
ALT0950	"Repair bus access modules (BAMS)" Changed from NT to T because this is not addressed on the LS
ALT1140	"Repair integrated platform management system (IPMS) console" NT to T because it is a required skill that is not trained elsewhere
ALT2050	"Troubleshoot onboard training system (OBTS)" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
ALT1210	"Repair Onboard training system (OBTS)" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
ALT0980	"Repair ~~CCTV~~ Server" *[*Note: CCTV removed by board decision 16 May]* NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
ALT1840	"Troubleshoot ~~CCTV~~ Server" *[*Note: CCTV removed by board decision 16 May]* NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
ALT2100	"Troubleshoot precision control equipment" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
ALT1280	"Repair precision control equipment" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
AKT0330	"Calibrate temperature sensors" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
HT0150	"Operate Utility Software for WAGO modules" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
HT0050	"Operate IPMS" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
HT0060	"Operate IPMS systems applications" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
ALT1930	"Troubleshoot fibre optic network" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
ALT0850	"Manage unclassified network" NT to T because it corresponds to previous IPMS training that is still required to be trained and not trained elsewhere
15 May 18	
Discussion with MS Div Cdr (LCdr Day)	Discussion about the requirements for all MS Electricians to be controls technicians so there is redundancy onboard ship (currently only one posted in each CPF). All participants agreed this is better than the current system with the single controls technician onboard.

Decision/Event	Discussion / Detail
	Discussion on the platform-specific material on the course and whether there should be short platform-specific modules following the non-specific course. Recommendation was sent to the OA to have platform-specific delta training created which would not be directly linked to training. OA did not agree and stated that all training, whether platform specific or not would be included in the USQ-MS Elec course.
	Discussion on the necessity of the MCDV OJT which is currently 6 weeks at the end of the Legacy QL6A. The OJT is not well laid out in the legacy QSP and is treated differently on each coast because it is a non-mandatory module (has a MITE but is not required for the NQual). The TDO suggested that MCDV specific training pieces should be provided as just in time training that could be a job aid or DL training that any electrician posted to an MCDV could take prior to their posting (no course required). Because the OA doesn't want platform specific modules, recommendation is to incorporate the MCDV theoretical elements but not the OJT portion in the USQ-MS Elec.
	Discussion on the gap that may present itself when the requirement for high voltage tech increases. Members agreed that training may need to be adjusted when high voltage requirements are better defined.
16 May 18	
Board is Convened	The TDO informed the members that the decision to turn the working group into a full board was made.
TDO Direction	The TDO briefed the board on what the purpose of the board is and where they will start. The board will start with the grouping and scalaring
MCDV specific tasks from legacy QL6A	"Advise senior staff on technical readiness" The board determined this was a NT because it is covered in RQP2
	"Control technical readiness" The board determined this was a NT because it is covered in RQP2
	"Conduct electrical systems trials" The board determined this was a NT because it is covered in RQP2
	"Maintain PLC" The board determined this was taught on the LS
	"Troubleshoot PLC" The board determined this was taught on the LS
	"Replace PLC" The board determined this was taught on the LS
	"Perform trails on PG supervisor control system" The board determined this was taught on the LS
	"Implement repair and maintenance instructions (ISSC / WEAF)" The board determined this task should be moved to the RQ-PO2 [*follow-up action item #1*]
ALT1930	The board decided to change the task from "Troubleshoot fibre optic network" to "Troubleshoot network" because there are multiple types of networks they will be troubleshooting. Differences between "Fibre optic network" and "cat 5 network" will be knowledge elements of that task. [*follow-up action item #2*]
ALT0980	The board changed "Repair CCTV server" to "repair server" as there are multiple servers potentially requiring repair. Differences between the types of servers will be knowledge elements of that task. [*follow-up action item #2*]
ALT1840	The board changed "Troubleshoot CCTV server" to "Troubleshoot Server" (same reason as above) [*follow-up action item #2*]
"Manage network administration"	The board decided there needed to be a new task called "Manage network administration" which is a main task for the controls tech. [*follow-up action item #2*]
ALT 0290	"Configure CANBUS network" Changed from NT (original board decision) back to train because MS will be configuring parts of the CANbus network that weren't covered in previous training.
17 May 18	
ALT0130	"Calibrate analog measuring test equipment" was decided by the board to be a NT because TEMMES is the one who calibrates it.

Decision/Event	Discussion / Detail
ALT0250	"Complete IPMS pre-sail checks" was decided by the board to be NT because the checklist is in SSOs and the individual tasks are trained throughout the course
"Determine impact to MCDV of constraints on operations"	Task was TRAIN during the MCDV OJT for QL6A. Board determined that it is important for the Chief Eng on an MCDV and should be ~~trained at the RQ~~ level for all MAR TECH. *[*Note: Board decided on 1 Jun not to train this task at any level]*
PO 004 Troubleshoot digital control systems	PIC portion of the course is currently being developed by the SME Tim Devoe. Further design for this PO will come based on what Mr. Devoe develops.
22 May 18	
"Manage Network Administration"	The board decided that this task needs to be added to the JTAR because it is a task that all Sr Electricians will be required to do as part of their job. [*follow-up action item #2*]
Job title "Senior Electrician"	Board decided on the job title for the USQ-MS Elec graduate will be "Senior Electrician" to fall in line with the MAR TECH Job Spec.
"Maintain" definition	The board decided the definition for "maintaining" a system will include "inspecting", "adjusting", "troubleshooting", and "repairing" the system therefore the "maintain" tasks are encompassing of the other four tasks.
PO 002 and 003 (Analog systems and Managing network) training limitations	Board discussed the trainer limitations on the West Coast versus the East Coast. Currently the East Coast is better equipped for this training as it has been conducted out there almost exclusively. Board recommends Tim Devoe, the networking SME who set up the trainer on the East Coast, is brought out to NFS(P) to configure the trainer in the same way. Instructors should learn how Mr. Devoe interfaces PIC tech with lab volt resources IOT bring the West Coast training on par. Ideally there should be 10 units configured identically on each coast. [*follow-up action item 3*]
23 May 18	
Target Population analysis	The board completed a target population analysis. This raised a question as to the annual throughput for the course. Board decided that it would likely be 32: 8 per course, 2 courses per coast per year. TDO emailed OA to confirm.
Training establishment analysis	The board completed a training establishment analysis. They discussed the differences on each coast, noting that East Coast was more equipped to deliver training as they have been conducting legacy training.
	Board discussed the amount of workstations present in each trainer that will be used for training. Current configuration has 8-12 workstations per trainer / classroom. Course will have a max capacity of 8 based on least amount of workstations in a trainer. If these trainers are expanded, course capacity should expand to meet the new requirements which will likely affect training time (if there are any assessments which are individual).
	Board discussed the validity of using the universal trainer to conduct some of the training. SME instructors currently work in Halifax therefore there is a possibility some lessons could be delivered by VTC in the universal trainer by the SMEs. Scheduling considerations would have to account for the time difference and practical exercises would not be monitored by the SMEs (separate instructor would be required in the trainer). Board agreed the best course of action would be to get civilian SME instructors hired for NFS(P) with same skills as those at NFS(A) (or out-source the training to Camosun / equivalent college). Universal classroom could be used in the interim if required however the lack of practical help from the SMEs could be detrimental. [*follow-up action item 4*]
OP Amps	The board noted that Op Amps are taught to the "understanding level" in the LS elec course. At the MS level, the trainee will be taught how to analyze and apply the theory that was learned in the LS. The board examined psychomotor taxonomy to come to this conclusion
USQ-LS Elec review	While reviewing the LS Elec QSP to ensure none of the tasks are being re-instructed at the MS level, the board (namely the IPMS SMEs) noted that some tasks included at the

Decision/Event	Discussion / Detail
	LS level are actually MS level tasks (involving passwords / more detailed networking knowledge).
	The CCTV networking task should be removed from the LS because maintenance of the server should be carried out by the senior electrician or above. [*follow-up action item 1*]
	"Perform a flash up of the OBTS" teaching point in the LS course should be removed from LS course because this task requires passwords and changes the console configuration (i.e. a lot of responsibility) so should be done at a higher level. [*follow-up action item 1*]
24 May 18	
IPMS Policy	The board identified a shortcoming where there is no policy governing IPMS. This includes what rank / position you get access to different SICs and privileges. They noted that there is no list of who gets what access and therefore system administrators have too much power which could create security issues.
	The board recommends a policy be developed that clearly defines who has what control and who is required to perform each duty associated with IPMS. [*follow-up action item 5*] Policy should include a standard HMI screen layout for all ships.
PO 001 "Analyze Circuit Fundamentals"	Originally the content from PO 001 was to be included as an EO under "Troubleshooting Analog Control Systems" but the board decided that it should be included as its own PO because it also included information for digital control systems (not just analog).
	Content for PO 001 was taken directly from the existing PO 010 of the legacy QL6A. It is not based on any tasks on the JTAR but is required theory which is the foundation for multiple tasks that the Sr Electrician will be required to perform on the job.
	PO 001 shall be the first PO instructed on the course as it is required to be able to perform tasks for multiple other POs (primarily PO 002, PO 004).
	Board decided there was no need for a PC 001 because this will be summatively assessed during PO 002 and PO 004 as the trainees will need to use the theory learned in PO 001 to complete the assigned tasks in those POs.
EO completion	The TDO instructed the board on what is expected to be include in each EO. The board completed EOs that were then reviewed by the board and TDO to ensure that everyone was working to the same standard.
"Analyze" definition	Board defined the term "Analyze" equivalent to the JTAR: "to examine critically for the purpose of interpretation or explanation". They discussed that Analyze will generally include the tasks of "measuring", "comparing", and "evaluating".
EO 005.02	The board decided there would be no EC 005.02 because it will be formatively assessed throughout PO 005-009 as these tasks are steps to perform subsequent tasks in those POs.
PO 005-009	The board determined these POs must be taught in sequential order in the lesson order from Ch 3 and Annex G of this QSP; it will not be possible to deviate from that order as they are sequential tasks that build on each other.
Annual throughput	Email from Training Manager: - "Originally, we had two (2) x QL6A ETECH serials of 8 students in Halifax only because of the lack of Training Aids and Qualified Instructors to teach subj course at NFS(P). - When QSP for USQ-MS ELEC will be completed and endorsed by CNPTG, we will seek assistance from Camosun College to teach/instruct. - In the meantime, NPTG HQ will schedule USQ-MS ELECT only at NFS(A). - I recommend scheduling two (2) serials of 8 students per fiscal year (at NFS(A))."
25 May 18	
PC 002 "Troubleshooting analog control systems"	The board discussed the Performance Check for "Troubleshooting analog control systems". Currently, legacy training has a written test with small practical portion which is weighted lower than the written component. The TDO suggested that a more performance-oriented test may be more appropriate.
	The board stated that the complexity of electricity in the control systems makes it impractical to assess in a trainer and maintain realism. Though the LabVolt trainer is an

Decision/Event	Discussion / Detail
	effective tool to develop skills, it is simulating a system in the span of 10 feet, whereas on ship this distance can be over 300 feet and involve many more variables
	The TDO suggested a scenario based assessment ("choose your own adventure" Tactical Decision Game). The board stated that this is too complex and will be too time consuming to be an effective means of testing. They believe that every trainee would need to be run through each scenario individually, taking too much time and being too complex to be effective.
	The board stated that experience and shipboard problem-solving are the only ways of practically assessing this training, therefore without OJT, it would need to be assessed using a written / theory based exam. An instructor can see based on the trainees understanding of the theory if they have the capability to effectively troubleshoot the system. The exam should include scenarios and links to work on the job wherever practical.
EO 005.04 "Maintain PGSCS"	The board requested the help of PO1 Derek Blanche to advise on this EO as it is MCDV-centred and none of the board members have recent MCDV experience. PO1 Blanche, NTDC(P), is qualified as a MCDV Chief Engineer.
	PO1 Blanche stated that most of the training at this point on CAMS is OJT and is difficult to develop ECs for. He believes that it would be more beneficial to give them an introduction at this point with more to be learned through experience once posted to an MCDV.
	PO1 Blanche advised the board that there are civilian contracts (ISSC) that teach most of the CAMs information so going further in-depth on this training would be redundant.
	The board decided 005.04 will not have an EC because the practical application is similar to other platforms of ship (includes all the same components) and it's only the configuration which is different. MCDV platform specific theory will be taught to the students with a tour during this EO.
28 May 18	
"Maintain PGSCS"	Board decided this task would be better suited under "Troubleshoot Analog Control Systems" (PO 002) instead of "Manage Network Admin" (PO 005). It will fit under "Analyze Chopper Circuits" (EO 002.04)
Discussion with CPO1 Pryor (Occupation Manager)	
Job title	Confirmed as Sr Electrician.
MCDV material in the course	Discussed keeping platform specific material so that any MS or PO2 can be posted to an MCDV as a senior electrician. The Senior Electrician will not be a watchkeeper onboard an MCDV; they will always be in the role of Sr Elec in case of emergency.
MCDV CSE tech	Sr Elec is generally divisionally in charge of the CSE tech and can therefore be expected to know some of the CSE stuff. Decision was CSE training should be amended or reviewed to include MCDV specific material so the Sr Elec onboard an MCDV is not responsible for CSE problems. [*follow-up action item 1*]
Target population	Intent of the course is to give the training to a brand new MS or senior LS. Preferably this would come a few years following the USQ-LS course, and a few years prior to RQ-PO2.
IPMS Position onboard ship	All MS Elecs will have IPMS training. "We have always had designated positions on ship, so there easily can be two MS positions (or 1 PO2 and 1 MS) designated as IPMS techs." CPO1 Pryor is unsure whether or not the policy will be to have only one with administrative passwords / privileges or whether all MS (and above?) would have it to take away the one point of failure.
MAR TECH benefits	Benefit of having general engineering training at the OS/AB level is easier troubleshooting later on. When they go to troubleshoot a system, they'll have a better overall understanding of how that system works so will not need to have a mechanical engineer with them. They will see the bigger picture.
AOPV training	ICT coming from the contractors with little to no input from RCN training system (TDO input but not MAR TECH specific). Currently we will not be including AOPV specific training on these courses; potential amendments will come once ICT is complete.

Decision/Event	Discussion / Detail
High voltage training	Currently training in the USA. Safety is the main concern for high voltage; systems are basically the same.
TEL / trainers	Trainers should be configurable for multiple platforms. PCOT could have all platforms available so trainees can see the differences between platforms for the IPMS and equivalents.
	Lab support staff (civilians) are lacking on the West Coast which leads to a lack of continuity, trainers which are not configured properly, and instructors which are not experts with the material. Capture the need for SME instructor in the resources section of the QSP for approval.
Beamex	Board discussed incorporation of Beamex Calibration equipment. Board decided it should be at the LS level. [*follow-up action item 1*]
EO development	
HT0120 "Operate Shore Based Trainer"	Board decided this task is NT. Most of the training for this task is covered by "Operate OBTS" and "Operate Orchid IS Software"; anything specific to the shore-based trainer (not done onboard ship) is experiential for those personnel posted to NFS.
Theory ECs/PCs	Board discussed the reasoning behind having theory ECs versus practical ECs.
	Troubleshooting control systems (PO 002, 004): many of the ECs are theory-based with practical portions covered by labs. During the labs, the trainees require instructor sign-off before moving forward with the lessons, therefore these act as ECs. Trainees need to pass all labs to pass the EO. Labs reinforce the theory piece by piece while the written EC brings together all the concepts, ensuring the trainee has gained the required knowledge/capabilities.
	Troubleshooting consoles / devices (PO 006): Some ECs are divided into two parts (practical and theory). The practical portions assess the most difficult/common troubleshooting faults while the theory parts cover the faults and procedures which are not assessed practically. Practically assessing all trainees on all possible faults / procedures is too time and resource intensive. Given the importance placed on the steps involved in troubleshooting, expressing the procedure in written format is sufficient to prove that the trainees can do the job.
29 May 18	
Position title	Discussion on the position title for the technicians onboard ship. Currently they have an "IPMS tech" onboard, but IPMS is the system onboard a CPF and therefore is platform-specific. Board (with Chief Pryor) decided, at the MS elec level, there will be a "Shop MS" and "Controls tech" to keep it non-platform specific.
"Troubleshoot consoles / devices"	Originally troubleshooting RTU and LOP were included in one PO with WAGO and HMI. Board decided the scope of this PO was large and assessment / on job performance was complicated, therefore they divided the original "Troubleshoot consoles/Devices" into "PO 006 Troubleshoot Consoles / Devices" which includes WAGO, HMI; "PO 007 Troubleshoot RTU" and "PO 008 Troubleshoot LOP". This will allow for a summative assessment on each separately as they are different overall performance (completing one does not necessarily mean you can do the other).
	PC 006 will be a written exam because it can be detrimental to the equipment to have them do it practically since L3 didn't supply a troubleshooting platform (i.e. setting faults and changing setting multiple times in quick succession). The written exam should sufficiently show the trainees are capable of completing the required processes.
ALT0270	Board (with Chief Pryor) decided to change the task from "Configure b/w and colour printers" to "Configure printers". [*follow-up action item 2*]
ALT0940	Board (with Chief Pryor) decided to change the task from "Repair B/W and colour printer to "Repair Printers". [*follow-up action item 2*]
30 May 18	
ALT1040	Board (with Chief Pryor) decided to NT the task "Repair CRPP system" because you repair by replacing the cannon plug which is previously trained at AB level.
EC 009.01	Assessments from legacy training are fairly similar to the current requirements. They can be amended by developers as required to conform to the new structure.

Decision/Event	Discussion / Detail
31 May 18	
EC 003.04	Board decided "Maintaining Nodes" does not require an EC as the tasks contained are small and are touched on during other phases.
PC 003	Board decided "Manage unclassified network" would be assessed by a written exam that would incorporate visual diagrams and pictures that the trainees would have to identify and explain. It is not a practical assessment because the main task of managing a network is large scope and would be difficult to do in the trainer.
EC 007.02	Board decided "Analyze Core Software" will not be assessed because it is for knowledge. They are not allowed to make any changes because this could damage the system but it's important to know about the core software.
1 Jun	
RoP Review	
WEAF and ISSC	Board discussed where these teaching points should be added. They affect both Mech and Elec, therefore it would make sense to have them at an RQ level. Monthly Maintenance is on the same computers as WEAFs and is done differently as well, therefore should be included as a TPt. Board decided (with Chief Pryor) these points should be added to the RQ-PO2. [*follow-up action item 1*]
ALT2100 and ALT1280	"Troubleshoot precision control equipment", "Repair precision control equipment" Should be added as tasks in 002.02. They are included in most labs throughout the course
"Test Printed Circuit Cards"	Task from QL6A originally listed as Train by the board. The board decided this was NT
"Determine impact to MCDV of constraints on operations"	Task from legacy QL6A MCDV OJT. Board discussed whether this is required to be trained. By the PO2 level they will be able to brief command (trained at both watchkeeper levels). They don't often think in terms of "operations" but they brief based on technical considerations for command to consider constraints on operations. Board decided it was not a required task to be added to JTAR or trained.
IPMS Policy	Board discussed the lack of an IPMS policy and how this can be detrimental. CCTV use involves ethics and the use of footage should be standardized (discussion with JAG as to what is legal or not). Passwords and privileges for systems should be standardized. Security policy should be standardized; adding IPMS network to the network agreement (or confirming it does apply) would work well. Many of the ships have a COTM, however it should be fleet standard. Board realized creating policy is above their level, however the ethics of managing a network should be added to the course. Board decided to add an LO on ethics in EO 003.01 "Maintain Network"
Ch 1 and 2	
PLARs for legacy	Legacy qualified IPMS techs will need to PLAR legacy QL6A if loaded on the course.
	Legacy qualified IPMS techs will be PLAR'd for this course; anyone holding the qual will not be required to do this course. They will still have to submit a PLAR, however it will be easily granted. They will not hold the MCDV portion of training.
Modularization	Board discussed modularizing the course, separated between analog and digital controls and distributed control platform management system. PO 001-004 would be one module; PO 005-009 would be second module. Because QL6A legacy qualified personnel would not have IPMS training, they would only have to do the second module. They could also be PLAR'd just for several POs without the modularization.
	Modularization could benefit scheduling: if candidates were not able to complete the full course for operational reasons, they could complete it by module.
	It's unlikely that modularization would benefit candidates in the long-term.
	Board (with Chief Pryor) decided the benefits are not substantial enough to warrant modularization, therefore the course will not be modularized at this time.
Course capacity	Maximum based on trainers (only 8 workstations in East Coast controls classroom); Minimum based on resource management (waste to train any less than 4); Optimum based on resource efficiency and group dynamic.
Prerequisites	Board discussed the requirement for the USQ-MS Elec qualification to get the RQ-PO2 training. Currently the RQ-PO2 QSP states "USQ-MS (mech or elec)" as a pre-requisite,

Decision/Event	Discussion / Detail
	however there is no USQ-MS Mech. This course will still be a requirement for the RQ-PO2. [*follow-up action item 1*]
Review	
Trainer maintenance	Board discussed the maintenance of the labs and trainers. On the East Coast, there is a maintenance team whose job is to maintain the labs; West Coast has unqualified MAR TECHs who have been attach posted in to the school but are unable to instruct. Maintenance increases the life-span for trainers and ensures the candidates have the most realistic training possible (without trainer glitches). Board recommends NFS(P) and NFS(A) develops a plan to standardize training resources and support for MAR TECH training aids. This could include shipping West Coast items to the East coast for maintenance, or sending maintainers from the East Coast to West Coast when required. Currently there is a plan to hire a training aid manager on the West Coast to match the East coast. [*follow-up action item 3 and 4*]
EC 003.01 and PC 003	EC for "Maintain Network" and PC for "Manage unclassified network" will be a written assessment to save time and resources. It would take too much time to test all trainees on all possible networking troubleshooting scenarios and they will be formatively assessed throughout on labs. Also, the lab in the school will be set up differently than the network onboard ship.

Guidance to Instructional / Development Staff:

1. Annex M is included with details on the boards vision of target population and current training establishment capabilities for use by developers as required.
2. Training material from the legacy ETech RQ-MS course (for PO 001, 002, 004) and legacy IPMS Technician course (for PO 003, 005-009) can be used to develop the course material for this training. The material should be up-dated to conform to current NTDC(P) standards and to include any additional teaching points / tasks as required.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	QSP changes actioned for additions or subtractions in other QSPs (USQ-LS Elec and RQ-PO2) that were noted during board (IAW RoP discussions)	NTDC(P) MS Div	15 Jun 18
2	SPQR actioned changing and adding tasks to the JTAR (IAW RoP discussions)	NTDC(P) MS Div	29 Jun 18
3	NFS(P) labs and training aids updated to be in line with NFS(A) labs and training aids	NFS(P) MS Div	15 Dec 18
4	SME instructor assigned at NFS(P) with equivalent skills/qualifications as those at NFS(A) to align bi-coastal training	NFS(P) MS Div	15 Dec 18
5	IPMS Policy developed	IPMS PM	15 Dec 18

ANNEX D - *QSPWB SCALAR*

The mapping of the task analysis from the QSPWB is detailed in the standard paragraphs of the POs and the mapping of the instructional analysis from the QSPWB is detailed in the standard paragraphs of the EOs in Chapter 3 of this QSP. A scalar can be found in the embedded document.

Annex D_Scalar_v3.pptx

MARPAC RDIMS #827901
Chap 1 approved 17 06 2022
Chap 2, 3 approved DD MM YYYY

ANNEX E - *ON JOB PERFORMANCE RECORD (OJPR)*

The USQ-MS for Senior Electrical Specialists does not include an OJT component therefore this Annex is left intentionally blank.

ANNEX F - *ACRONYMS AND ABBREVIATIONS*

Acronym/ Abbreviation	Term or definition
Generic Training Terms	
ATT	Actual Training Time
BAT	Base Administrative Time
CCA	Curriculum Control Authority
EC	Enabling Check
EO	Enabling Objective
IAW	In accordance with
min	minutes
MITE	Military individual training and education
NPTG	Naval Personnel and Training Group
NQUAL	National Qualification
PC	Performance Check
PO	Performance Objective
PT	Physical Training
QSP	Qualification Standard and Plan
RCN	Royal Canadian Navy
RQ	Rank Qualification
TAT	Training Administrative Time
TE	Training Establishment
TRB	Training Review Board
TTT	Training Travel Time
USQ	Unique Specialty Qualification
Senior Electrical Specialist Specific Terms	
ADAM	Analog Digital Acquisition Module
ADT	Application downloader tool
ASCII	American Standard Code for Information Interchange
BAM	Bus Access Module
BDCS	Battle Damage Control System
BITE	Built-in Test Equipment
CAMS	Central Alarm Monitoring System
CANbus	Controller Area Network bus
CBRN	Chemical, biological, Radiological, Nuclear
CCA	Circuit Card Assembly
CCTV	Closed Circuit Television

COM	Communication
CPU	Central Processing Unit
CRPP	Controllable Reversible Pitch Propeller
DC	Direct Current
DCC	Damage Control Console
EHM	Equipment Health Monitoring
FMF	Fleet Maintenance Facility
FSR	Field Service Representative
GMD	Graphical MCS Debug
HLF	History Log File
HMI	Human Machine Interface
HRC	Halon Release Controller
IFDMS	Integrated Fire and Damage Monitoring System
IP	Internet Protocol
IPMS	Integrated Platform Management System
IS	Instructor Station
LAN	Local Area Network
LCP	Local Control Panel
LOP	Local Operating Panel
LSU	Local Scanning Unit (MCDV)
MCC	Machinery Control Console
MCDV	Maritime Coastal Defence Vessel
MCRT	Machinery Control Room Trainer
OBTS	On Board Training System
OSI	Open System Interconnection
PCOT	Platform Control Operator Trainer
PDDB	Power Data Distribution Boxes
PGCSC	Power Generation Control Systems and Components
PGSCS	Power Generation Supervisory Control System
PIC	Programmable Intelligent Computer
PID	Proportional integral derivative
PLC	Programmable Logic Controller
POU	Portable operating units
PPE	Personal Protective Equipment
PROCMON	Process Monitor
RDA	Request Data from a Specific Address
RTD	Resistance Temperature Detectors
RTU	Remote Terminal Unit
SCR	Silicon controlled Rectifier
SFC	Static Frequency Converters

SKI	Shafts Knots indicator
TBC	Transmit Bit Clear
TBS	Transmit Bit Set
TDA	Transmit Data to a Specific Address
UPS	Uninterruptable Power Supply
VRTU	Versa Module Europe (VME) Remote Terminal Unit

ANNEX G - *SUGGESTED SEQUENCING AND TIMINGS*

1. This annex is to be used by Campus Support to properly schedule all required classrooms, trainers, ships, personnel, etc for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).
2. The tables are completed in the preferred sequence for lessons and resources. They are to be read in conjunction with the proposed course schedule (when available).

Annex G_Sequencing and timings_v3.xlsx

EO ###.##	EO Performance Statement	Personnel resources	Required Resources (for scheduling)	Trg Time per resource (mins)	EC Time (mins)	PC Time (mins)
PO 001: Interpret Circuit Fundamentals						
PO ATT:	**1840**	NOTES:				
EO 001.01	Interpret Digital Circuit fundamentals	Civ SME Instructor	Classroom	900	70	
EO 001.02	Interpret Digital Circuit fundamentals	Civ SME Instructor	Classroom	800	70	
PO 002: Troubleshoot analog control systems						
PO ATT:	**7030**	NOTES:				
EO 002.01	Analyze elements of controls	Civ SME Instructor	Classroom	1150	60	
EO 002.02	Analyze control circuits	Civ SME Instructor	Classroom	100		
			Lab Volt	100		

			Classroom	300		
			Lab Volt	300		
			Classroom	100		
			Lab Volt	100		
			Classroom	300		
			Lab Volt	150		
			Classroom	150		
			Lab Volt	75		
			Classroom	75		
			Lab Volt	150		
			Classroom		120	
EO 002.03	Analyze four quadrant control	Civ SME Instructor	Classroom	450		
			Lab Volt	300		
			Classroom		60	
EO 002.04	Analyze chopper circuits	Civ SME Instructor	Classroom	250		
			Lab Volt	300		
			Classroom	260		
		Civ SME Instructor MCDV Sr Electrician	MCDV electrical tour	400		
		Civ SME Instructor	Classroom		60	
EO 002.05	Analyze inverter circuits	Civ SME Instructor	Classroom	450		
			Lab Volt	300		
			Classroom		60	110
			Lab Volt			800
PO 003: Manage unclassified networks						
PO ATT:	**3740**	NOTES:				

EO 003.01	Maintain Network	JAG SME	Classroom	60		
		Civ SME Instructor	Classroom	740		
			LAN Lab	1000		
			Classroom		60	
EO 003.02	Maintain switches	Civ SME Instructor	LAN Lab	100	280	
EO 003.03	Maintain servers	Civ SME Instructor	LAN Lab	500		
			CCTV server lab and PCOT	50		
			PCOT		480	
EO 003.04	Maintain nodes	Civ SME Instructor	Classroom	130		
			LAN Lab	280		
			Classroom			60
PO 004: Troubleshoot digital control systems						
PO ATT:	**7260**	NOTES: Lessons are divided between classroom and lab time therefore preferable if lab is attached or near the classroom to cut down on travel time between stations.				
EO 004.01	Program microcontrollers	Civ SME Instructor	Classroom	700	60	
EO 004.02	Utilize digital control for electric motors	Civ SME Instructor	Classroom	750		
			Lab	150		
			Classroom	300		
			Lab	450		
			Classroom	50		
			Lab	50		
			Classroom	100		
			Lab	150		
			Classroom	250		
			Lab	350		

			Classroom		60	
EO 004.03	Analyze digital data	Civ SME Instructor	Classroom	750		
			Lab	150		
			Classroom	300		
			Lab	450		
			Classroom		60	
EO 004.04	Analyze open loop digital control using packets	Civ SME Instructor	Classroom	250		
			Lab	250		
			Classroom	90		
			Lab with system simulators	960		
			Classroom		60	
			Lab with system simulators			520
PO 005: Manage Network Administration						
PO ATT:	**1085**	NOTES:				
EO 005.01	Operate IPMS system applications	Instructor	PCOT	60	25	
EO 005.02	Operate IPMS software utilities	Instructor	PCOT	140		
EO 005.03	Analyze IPMS EHM data	Instructor	PCOT	100	320	
EO 005.04	Operate OBTS	Instructor	PCOT	50		
		Instructor	PCOT	50		
		FSR				
		Instructor	PCOT	100	200	40
PO 006: Troubleshoot consoles and devices						

PO ATT:	**1010**	NOTES:				
EO 006.01	Repair IPMS console	Instructor	PCOT	200		
			IPMS Trainer	180		
			PCOT		35	
EO 006.02	Operate utility software for WAGO modules	Instructor	IPMS Trainer	260	280	
			Classroom			55
PO 007: Troubleshoot RTUs						
PO ATT:	**4150**	NOTES:				
EO 007.01	Repair RTU	Instructor	PCOT	100		
			IPMS Trainer	3450		
			Classroom		50	
			IPMS Trainer			400
EO 007.02	Analyze core software	Instructor	IPMS Trainer	150		
PO 008: Troubleshoot LOP						
PO ATT:	**2810**	NOTES:				
EO 008.01	Repair LOP	Instructor	IPMS Trainer	360	200	
EO 008.02	Operate utility software for ADAM	Instructor	IPMS Trainer	75		
			Classroom	75		
			IPMS Trainer	1650		
			Classroom		50	
			IPMS Trainer			400
PO 009: Configure CANbus network						
PO ATT:	**2940**	NOTES: Lessons are divided between classroom and lab time therefore preferable if lab is attached or near the classroom to cut down on travel time between stations.				
EO 009.02	Configure PDDB	Instructor	Classroom	100		

			IPMS trainer	100		
			Classroom		50	
EO 009.01	Configure BAMs	Instructor	Classroom	100		
			IPMS Trainer	100		
			Classroom	100		
			IPMS trainer	1600		
			PCOT	300		
			Classroom		50	
			IPMS Trainer			440
		Total EO time:	**26220**		**ATT:**	**31865**
Total EC time:	**2820**	**Total PC time:**	**2825**		**Total mins:**	**36884.75**
PT:	**2867.85**	**Course admin time:**	**2151.9**		**Total days:**	**92.2**

ANNEX H - *ADDITIONAL TRAINING RESOURCES*

In accordance with Chapter 1 para 3-4, add any extra tables or paragraphs used for training resources in this annex. (**Annex H not to be deleted**)

Stationery given at start of course (per trainee)	• Pencil • Pen • Highlighter • Eraser • 200 sheets lined paper • 50 sheets graph paper • 2.5” binder
General classroom to contain, as a minimum	• 1 DWAN computer for instructor • 1 x projector • 2 x 8x4 whiteboards • 8 multicoloured whiteboard markers • Whiteboard eraser
Loaned textbooks (PO 001-004)	• Deal Richard, Cisco Certified Network Associate, McGraw Hill • Bateson Robert, Introduction to Control System Technology, 3rd Edition • Introductory Electronic Devices and Circuits, 6th Edition, (Paynter, Robert • Basic Technical Mathematics with Calculus, 8th Edition (Washington, Allyn J.) • Network+ Study Guide 3rd Ed (Goth, David)
Test Equipment	• TRUE RMS MULTIMETER Fluke 112 • TRUE RMS MULTIMETER Fluke 115 • TRUE RMS MULTIMETER – REMOTE DISPLAY Fluke 233 • MULTIMETER Fluke 8020A • INSULATION /MEGO METER Fluke 1587 • INSULATION /MEGO METER Fluke 1507 • INSULATION /MEGO METER Meter Master • INSULATION /MEGO METER Yokogawa MY40 • INSULATION /MEGO METER Megger BM10 • TRUE RMS CLAMP METER – REMOTE DISPLAY Fluke 381 • IR THERMOMETER -MINI Fluke 62 • CLAMP METER GREENLEE / CM-600 • CLAMP METER GEN ELECTRIC • DIGITAL OSCILLOSCOPE TDS 210

	• RESISTANCE DECADE BOX TENMA 72 7220 • TACHOMETER HASLER • TACHOMETER TENMA 72-7270 • TACHOMETER- HAND ASAHI • TACHOMETER – DIGITAL Monarch PT99 • CORROSION POTENTIAL METER Electrocatalytic Inc • PRESSURE / VACUUM MODULE Fluke PV 350 • STROBOTAC (STROBOSCOPE) Gen Radio Comp Type1531-A • SPEED INDICATOR ELLIOT • REFRACTOMETER MISCO 7064VP • BREADBOARD SK-10 E & L INSTRUMENTS
LabVolt equipment	• Accessory Bay LabVolt • Power Supply 8821-20 • Power Supply 8840-00 • DC Machine 8211-00 • Dynomometer 8960-10 • Synchronous Machine 8241-00 • Component Panel 9127-10 • Contactor 9106-20 • Control Relay 9108-10 • DC Volt/Ammeter 8412-10 • AC Voltmeter 8426-00 • Power Diodes 8842-10 • Smoothing Inductors 8325-10 • Power MOSFETs 8837-00 • Chopper Controller 9092-00 • Thyristor Firing Unit 9030-30 • Current/Voltage Isolator 9056-10 • Power Thyristors 8841-20 • Resistive Load 8311-00 • Inductive Load 8321-00 • Data Acquisition 9063-00 • Tachometer 8931-00 • Control Bay 8846 • DC Motor/Generator 9318-00
LAN / Network lab	• Switches • Servers • Nodes • Cat 5 cables • Fibre optic cables
PIC trainer accessories	• PIC Trainer Board • Analog temp Sensor

	• Digital temp sensor Maxim • Ceramic Resonator • LCD Sub Module • Graphic Sub Module • PICKit-3 Programmer • RS485/422 Board • Terminal Block • Connector Pins • Connector • Quad Switch • Knob
Tool Kit	• Digital logic probe; • Wire stripper; • Jeweler screwdriver; • Needle-nose pliers; • Multi-bit screwdriver • Wire cutters
IPMS Maintenance Trainer	• WAGO • RTU • LOP • ADAM • BAM loop Network • CCTV • Various Sensors • CAT 5 cables and terminations • Network Test Equipment • Fault insertion controller • Soldering Kit
PCOT	• IPMS Computer Network with Pre-built scenario's • EHM Server Database • 10 Computer Stations • Instructors computer and Projector • 2x Whiteboards

ANNEX I - *COSTING INFORMATION*

Include information about cost in whichever format is applicable (**Not to be deleted**).

Template Annex I Costing.xls

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: TDL 300		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX J - *COURSE HISTORY*

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

<table>
<tr><th>Reason for amendment (with date)</th><th>Details on course history</th></tr>
<tr><td rowspan="2">QSPWB
Apr-Jun 2018</td><td>The USQ-MS Electrical came about as a result of the MAR TECH trade creation. It is a combination of legacy IPMS Maintainer and RQ-MS Electrical Technician training. Many of the teaching points have remained the same, however the structure is amended to reflect the amalgamation and the format updated to the West Coast QSP pilot template.</td></tr>
<tr><td>The original QSPWB was convened in April 2018 however after the board realized most of the training was already developed, the board moved on to the USQ-PO2 Elec instead. A small review board with a TDO and 2 SMEs put together a plan for the course (i.e. how the two legacy courses would be combined), including going through the Train/No Train list from the disbanded board as well as the task lists for the two legacy courses. It was recommended to the division by the TDO to conduct a full QSPWB given the size of the project, therefore a second QSPWB was convened in May 2018 (see Annex C for detailed discussions / decisions from these three boards).</td></tr>
<tr><td rowspan="4">QSP QA review
Dec 2021 - Feb 2022</td><td>A QSP QA check was conducted to see if the QSP was ready to be approved and made steady-state.
The QSP was reviewed by a SME (PO2 Brad Salmon) and TDOs for completeness, accuracy, formatting, consistency, etc. IAW the QSP QA Checklist.
QSP QA Checklist - USQ MS Snr Elec Spe</td></tr>
<tr><td>During this review, it was confirmed by the TM the course title and codes are:
MARPAC-#1214097-v1-MAR_TECH_MITE_ar
<table><tr><td>USQ-MS SENIOR ELECTRICAL SPECIALIST</td><td>122021</td><td>ALUZ</td></tr></table>The original title of “Senior Electrician” was therefore changed throughout the document including the title page, Ch. 1, POs, and anywhere within EOs or ECs where it is listed. The MCDV Sr Electrician title was kept with no change as it is specific to a class of ship. No changes to the job title were made in the RoP or Annex B as these were created specifically during the QSP WB where “Sr Electrician” was the current title.
Other code that was confirmed and updated through Ch 1:
USQ-S1 ELECTRICAL SPECIALIST 122020 ALUT
MAR TECH RQ-PO1 (ALUV)</td></tr>
<tr><td>Course duration was updated to reflect all 19 weeks. Originally the BAT, TAT, and PT calculations were based on the total training time in weeks, but this does not account for the added 2 weeks caused by the admin and PT requirements. This increased the overall course duration from 93 days to 95 days but should improve scheduling. This will be re-evaluated during the pilot.</td></tr>
</table>

	PO and EO standards were compared to make sure required tasks were captured and wording was equivalent. Some PO and EO standards are not in the same order because POs are in the order of performance on the job while Eos are in the order of instruction; this reflects what is required based on CFITES structures.
June 2024	Prerequisite for USQ Engineering Watchkeeper removed.

ANNEX K - *ASSESSMENT PLAN*

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 001.01	EO 001.01	Analyze digital circuit fundamentals	Written assessment for assessor feedback based on standardized solutions.	70%	Remedial Training / Re-test
EC 001.02	EO 001.02	Analyze analog circuit fundamentals	Written assessment for assessor feedback based on standardized solutions.	70%	Remedial Training / Re-test
EC 002.01	EO 002.01	Analyze elements of control	Written assessment for assessor feedback based on standardized solutions.	70%	Remedial Training / Re-test
EC 002.02	EO 002.02	Analyze control circuits	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	70%; Pass all labs	Remedial Training / Re-test
EC 002.03	EO 002.03	Analyze four quadrant control	Written assessment with emphasis on troubleshooting concepts for assessor feedback based on standardized solutions.	70%	Remedial Training / Re-test
EC 002.04	EO 002.04	Analyze chopper circuits	Written assessment with emphasis on troubleshooting concepts for assessor feedback based on standardized solutions.	70%	Remedial Training / Re-test
EC 002.05	EO 002.05	Analyze inverter circuits	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	70%; Pass all labs	Remedial Training / Re-test

PC 002	PO 002	Troubleshoot analog control systems	Part A: Written assessment for assessor feedback based on standardized solutions. Part B: Practical assessment in lab. Troubleshoot complex analog control systems	Part A: 70% Part B: SAT all checklist items	Remedial Training / Re-test / TRB after second unsuccessful attempt
EC 003.01	EO 003.01	Maintain networks	Written assessment with emphasis on troubleshooting concepts, for assessor feedback based on standardized solutions.	60%	Remedial Training / Re-test
EC 003.02	EO 003.02	Maintain switches	Practical assessment in LAN lab; Configure switches individually	SAT all checklist items	Remedial Training / Re-test
EC 003.03	EO 003.03	Maintain Servers	Practical assessment in PCOT; Manage EHM server database	SAT all checklist items	Remedial Training / Re-test
PC 003	PO 003	Manage unclassified Network	Written assessment with visual cues for assessor feedback based on standardized solutions.	60%	Remedial Training / Re-test / TRB after second unsuccessful attempt
EC 004.01	EO 004.01	Program microcontrollers	Written assessment for assessor feedback based on standardized solutions.	70%	Remedial Training / Re-test
EC 004.02	EO 004.02	Utilize Digital control for Electric Motors	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	70%; Pass all labs	Remedial Training / Re-test

EC 004.03	EO 004.03	Analyze digital data	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	70%; Pass all labs	Remedial Training / Re-test
EC 004.04	EO 004.04	Analyze open loop digital control using packets	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	70%; Pass all labs	Remedial Training / Re-test
PC 004	PO 004	Troubleshooting Digital Control Systems	Practical assessment in lab with system simulators; Repair 2 faults to lowest repairable unit	70%	Remedial Training / Re-test / TRB after second unsuccessful attempt
EC 005.01	EO 005.01	Operate IPMS System Applications	Practical assessment in PCOT; flash up the system and carry out a series of tasks within the system.	SAT on all checklist items	Remedial Training / Re-test
EC 005.03	EO 005.03	Analyze IPMS EHM Data	Practical assessment in PCOT; Based on scenario; Retrieve appropriate signals from EHM for analysis	SAT on all checklist items	Remedial Training / Re-test
EC 005.04	EO 005.04	Operate OBTS	Practical assessment in PCOT; start a pre-loaded scenario on HMI	SAT on all checklist items	Remedial Training / Re-test
PC 005	PO 005	Manage Network Administration	Practical assessment in PCOT; carry out series of tasks in IPMS with assessor confirming completion at each step	SAT on all checklist items	Remedial Training / Re-test / TRB after second unsuccessful attempt
EC 006.01	EO 006.01	Repair IPMS console	Practical assessment in PCOT; create and setup a pre-set configuration for HMI	SAT on all checklist items	Remedial Training / Re-test

EC 006.02	EO 006.02	Operate utility software for WAGO modules	Practical assessment in IPMS maintenance trainer; operate WAGO software	SAT all checklist items	Remedial Training / Re-test
PC 006	PO 006	Troubleshoot consoles and devices	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	60%; Pass all labs	Remedial Training / Re-test / TRB after second unsuccessful attempt
EC 007.01	EO 007.01	Troubleshoot RTU	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	60%; Pass all labs	Remedial Training / Re-test
PC 007	PO 007	Repair RTU	Practical assessment in IPMS maintenance trainer; troubleshoot a fault in the RTU	SAT all critical, 70% SAT remainder checklist	Remedial Training / Re-test / TRB after second unsuccessful attempt
EC 008.01	EO 008.01	Configure ADAM	Practical assessment in IPMS maintenance trainer; Configure an ADAM for use	SAT all checklist items	Remedial Training / Re-test
EC 008.02	EO 008.02	Troubleshoot LOP	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	60%; Pass all labs	Remedial Training / Re-test
PC 008	PO 008	Repair LOP	Practical assessment in IPMS maintenance trainer; troubleshoot a fault in the LOP	SAT all critical, 70% SAT remainder checklist	Remedial Training / Re-test / TRB after second unsuccessful attempt

EC 009.01	EO 009.01	Configure PDDB	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	60%; Pass all labs	Remedial Training / Re-test
EC 009.02	EO 009.02	Configure BAM	Written assessment for assessor feedback based on standardized solutions. Practically assessed during labs	60%; Pass all labs	Remedial Training / Re-test
PC 009	PO 009	Configure CANbus Network	Practical assessment in IPMS maintenance trainer; troubleshoot a fault in the BAM loop network	SAT all critical, 70% SAT remainder checklist	Remedial Training / Re-test / TRB after second unsuccessful attempt

ANNEX L - *TEL PLAN*

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta
PO 001 – 004	Gas turbine and alternator lab PIC trainer properly configured with RTU modules on the West Coast (based on East Coast model). OR PIC training at Camosun College	Available on the East Coast; limited components are on the West Coast but they are not integrated	Continuity on both coasts allowing training to be bicoastal which reduces TD costs and time away from home.
EO 003.03	CCTV server and cameras network with cat 5 cables and access node (one configuration for each coast).	Powerpoint / F2F lecture (not available on either coast)	Trainees have hands on experience with a CCTV network. Ability to troubleshoot and input faults without concern for affecting shipboard systems.
PO 005	Create EHM sandbox within the PCOT on both coasts to appear as live data	FSR load historic ship's database to West Coast PCOT (nothing on east Coast)	No requirement for FSR to input data. Not currently available on the East Coast. More realistic training since the trainees can search for current dates instead of based on historic data. Real-time tracking of EHM data to reflect performance on the job.
PO 005-009	PCOT able to emulate multiple platform configurations	PCOT only able to emulate CPF IPMS configuration; shipboard familiarization with equipment	More realistic training for all platforms without affecting shipboard operations. Ability to troubleshoot a larger variety of systems to make postings to different platforms smoother.

ANNEX M - *TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS*

The following information was compiled during the QSPWB (May 2018) to give context for developers. It can be removed from the QSP if desired after initial course development.

Key Target Population Details – MAR TECH USQ-MS Sr Elec Spec candidates

"The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners" -CFITES Vol 4

Characteristic	Target population details
General data	
Course capacity	8 Max based on East Coast PIC trainer capacity; 8 optimum
Expected annual throughput	2 course per coast each year = 32 total
Trade(s)	MAR TECH Electrical; Civilian control technicians / instrumentation mechanics possible (PLARs)
location(s) of the intended learners	Bi-coastal
Access to computers (on the job)	Yes
Average age	25-30
Learning skills and preferences	
Competencies (general)	Critical thinking; mechanical aptitude; technological orientation; problem-solving
Reading and writing level	Reading – above average for technical documentation Writing – average
Preferences / attitudes towards specific type of learning material and equipment (i.e. TEL, aversion to lectures)	Embraced technology = TEL oriented Independent working Hand's-on learning

Characteristic	Target population details
Educational level	High school + Technician courses
Previous military training experience	PLQ (generally) Up to USQ-LS Elec Some watchkeeper qualified
Current subject-matter competence	
Experience with target performance	Exposure to everything (USQ LS Elec)
Ability to perform any part of the POs	Simple aspects they can do
Knowledge of the subject matter (and how it was acquired)	Basics developed at the LS, refined here at the MS level
Perception of the impact of achieving the POs on self, work or career	High motivation: required for trade progression (everyone on the course is looking to advance). Gives them several years before promotion where they will be able to refine their skills on the job. In demand in Navy and civilian side
Job/career stage	Senior LS to newly-promoted MS = mid-career; 6-10 years in uniform. Not taken directly before a promotion
Affective Characteristics	
Interests	Ability to control power plan; automation
Attitude towards subject matter/ learning (academic self-concept)	Ready for time ashore; looking for mental stimulation / challenge
Anxiety level	Bi-coastal: happy to be at home Academic anxiety
Social Characteristics	
Relationship to peers	Familiar with peers; good relationships
Feelings toward authority	No issue
Tendencies toward cooperation or competition	Competition that breeds cooperation

Characteristic	Target population details
Other	
Fitness level	All pass FORCE test; not required Mental fitness: variable depending on previous op tempo; better if bi-coastal (at home)
Health status	Generally healthy without multiple appointments so they don't miss much class. Mostly sedentary classwork.

<u>Describing the learning environment</u> – NFS MS division

The "learning system" is composed of all factors that affect and are affected by the learning that takes place: learners (Target Population), instructional materials, the instructors, instructional equipment (such as computers, projectors, LMS, etc), the instructional facilities, and the community or organization.

1. What are the characteristics of the teachers/trainers who will be using these materials?
<u>Instructors on East Coast</u>: 2 civilians with extensive networking experience on extended contract; military instructors <u>Instructors on West Coast</u>: Military instructors currently; parts should be instructed by networking expert (not currently on staff)
What are the interests and preferences of the instructors? Civilians: networking is their specialty Military: IPMS instructors are picked based on qualifications and desire therefore it is their specialty
How do instructors feel about having instruction delivered via media or other non-traditional methods? Embrace it; use trainers and labs; lots of hands on and TEL
Do instructors like to use technology-based instruction? Yes; currently existing for legacy training
What experience have instructors had with technology-based instruction? Civilians: developed it Military: currently using it; trained on it
2. Are there existing courses/programs into which this piece of instruction must fit?

If so, what is the strategy used in these materials? Merging two legacy courses. Must fit into IPMS control systems on the ship
3. What hardware is commonly available in the potential learning environments?
Are computer workstations available? Classroom: West coast = 12 DWAN workstations; East Coast = networked (not DWAN); 8 workstations IPMS Trainer (both coasts): 10-12 workstations PIC trainer (East): 8 student workstations PCOT (Both coasts): 12 workstations Lab Volt (both coasts; West coast isn't configured properly yet): 12 workstations MCDV Trainer at NFS(Q): 12 Workstations AOPV trainer expected on each coast
4. What are the characteristics of the classes and facilities that will use the new instruction?
High tech setup; modernized in terms of tech